



Stony Brook Institute
at Anhui University

安徽大学纽约石溪学院



Chapter 5

THE NETWORK LAYER

Contents



1. Introduction
2. The Physical Layer
3. The Data Link Layer
4. The Medium Access Control Sub-Layer
5. **The Network Layer**
6. The Transport Layer
7. The Application Layer
8. Network Security

Chapter 7 Application	
Chapter 6 Transport	
Chapter 5 Network	
Chapter 3 LLC	Data Link
Chapter 4 MAC	
Chapter 2 Physical	

Contents



1. Network Layer Design Issues
2. Routing Algorithms
3. Congestion Control Algorithms
4. Quality of Service
5. Internetworking
6. The Network Layer in **the** Internet

5.1 Network Layer Design Issues



- Store-and-Forward Packet Switching
- Services Provided to the Transport Layer
- Implementation of Connectionless Service
- Implementation of Connection-Oriented Service
- Comparison of Virtual-Circuit and Datagram Networks

5.1.1 Store-and-Forward Packet Switching



- ISP(Internet Service Provider) 's Equipment
 - ✦ Routers
 - ✦ Transmission Lines
- Customers' Equipment
- DSL Modem

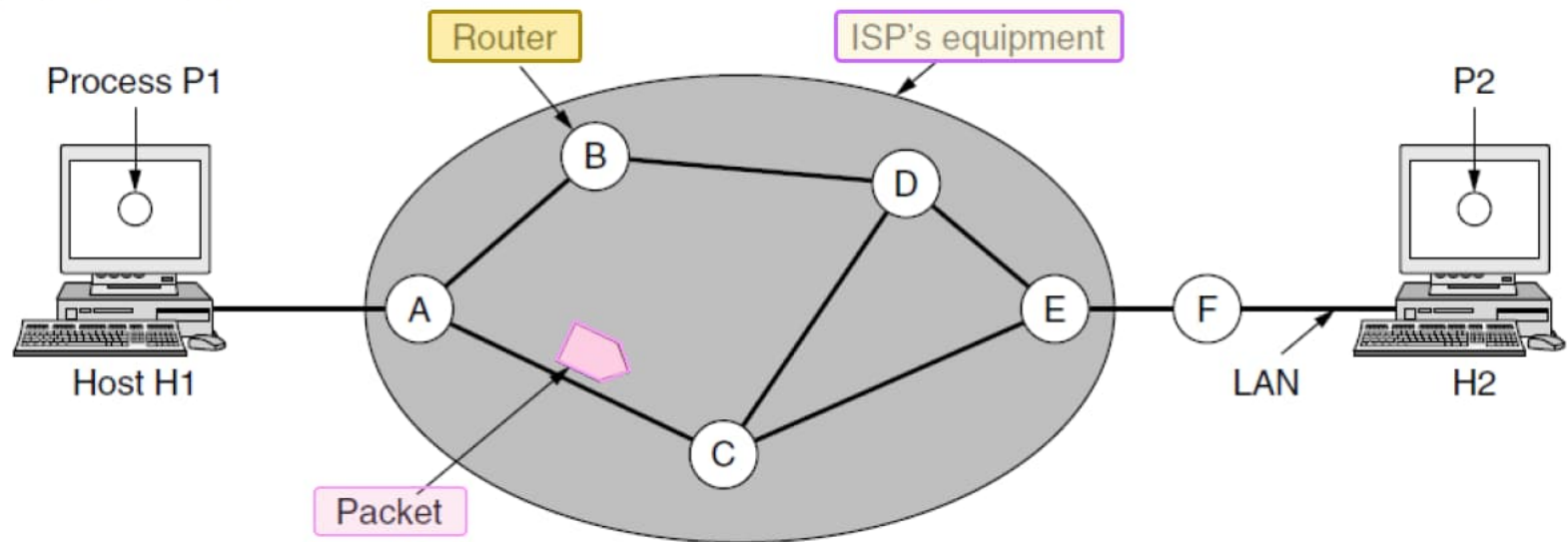


Figure 5-1. The environment of the network layer protocols.

5.1.2 Services Provided to the Transport Layer

- Network Layer / Transport Layer Interface
- Goals of services
 - ✦ The services are independent of the router technology
 - ✦ The transport layer should be shielded from the routers
 - ✦ Uniform numbering plan
- Battle Between Two Factions
 - ✦ Network service should be
 - Connectionless. (ARPANET IP)
 - v.s
 - Connection-Oriented. (X.25, FR, ATM, MPLS)

faction 派系

5.1.3 Implementation of Connectionless Service



➤ Routing algorithm

- ✦ manages the routing tables
- ✦ makes the routing decisions.

➤ Connectionless service

- ✦ packets are routed independently
- ✦ the packets are frequently called datagrams (in analogy with telegrams)
- ✦ the network is called a datagram network.

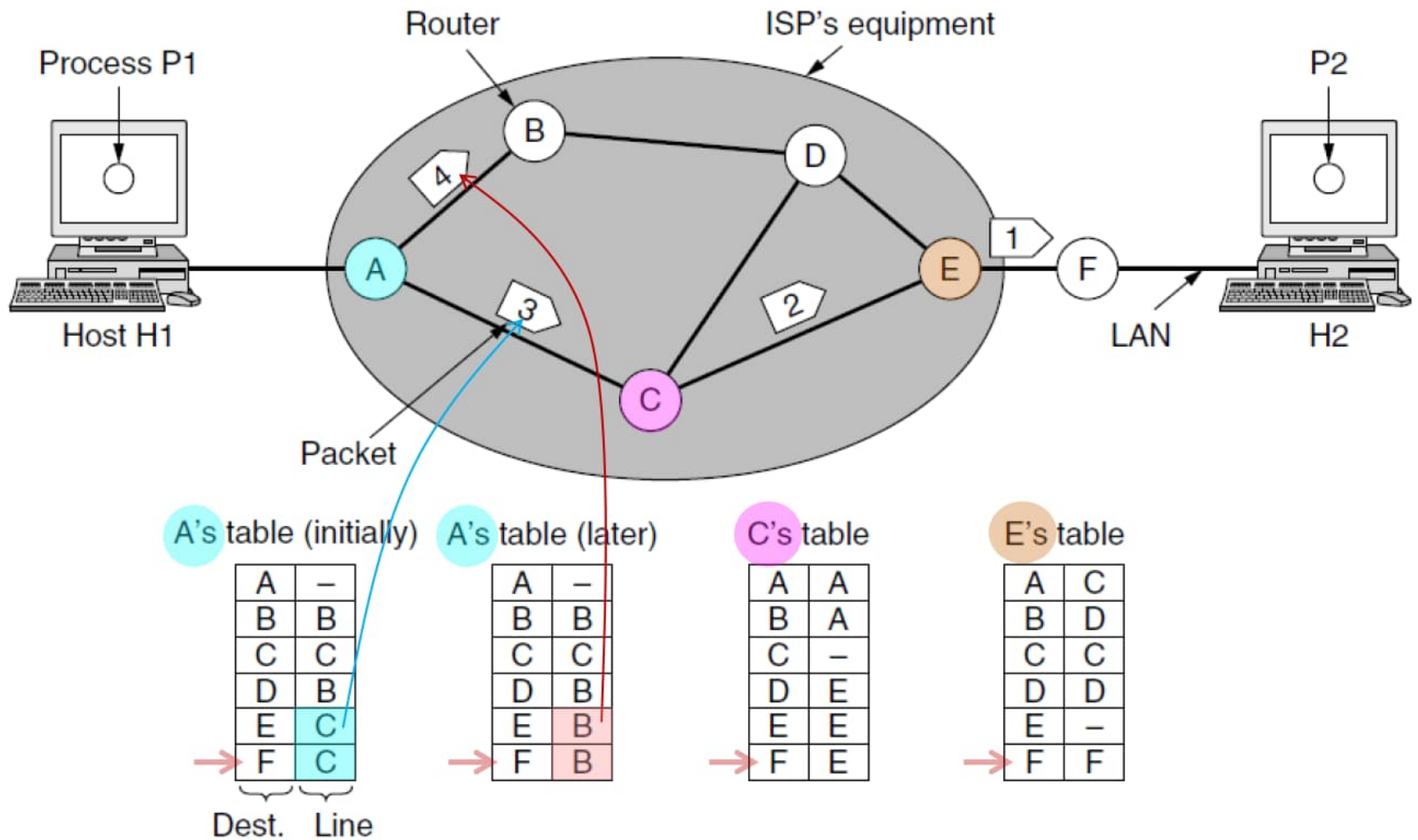


Figure 5-2. Routing within a datagram network.

5.1.4 Implementation of Connection-Oriented Service



- Connection-oriented service
 - ✦ a path from the source router to the destination router must be established before any data packets can be sent.
 - this connection is called a VC (virtual circuit)
 - the network is called a virtual-circuit network

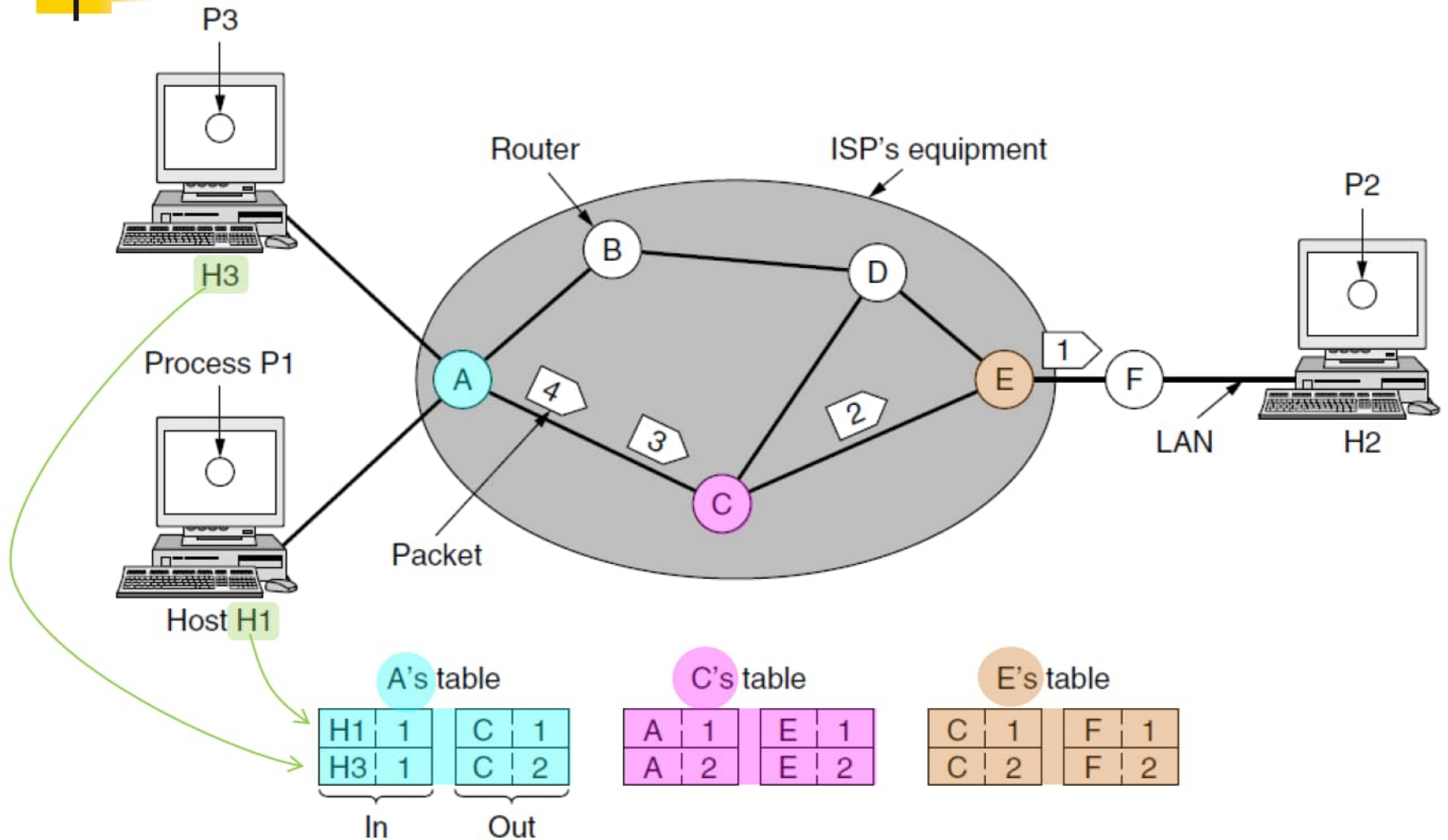


Figure 5-3. Routing within a virtual-circuit network.

5.1.5 Comparison of Datagram Networks and Virtual-Circuit

Issue	Datagram network	Virtual-circuit network
Circuit setup	Not needed ✓	Required ☒
Addressing	Each packet contains the full source and destination address ☒	Each packet contains a short VC number ✓ (cut-through switching)
State information	Routers do not hold state information about connections ✓	Each VC requires router table space per connection ☒
Routing	Each packet is routed independently ☒	Route chosen when VC is set up; all packets follow it ✓ (delivery in order)
Effect of router failures	None, except for packets lost during the crash ✓	All VCs that passed through the failed router are terminated ☒
Quality of Service	Difficult ☒	Easy if enough resources can be allocated in advance for each VC ✓
Congestion control	Difficult ☒	Easy if enough resources can be allocated in advance for each VC ✓

Figure 5-4. Comparison of datagram and virtual-circuit networks.

5.2 Routing Algorithms



- The **main function** of the network layer
 - ✦ is **routing packets** from the source machine to the destination machine.
- The routing algorithm
 - ✦ is that part of the network layer software **responsible for** deciding **which output line** an incoming packet should be transmitted on.
 - ✦ If the network uses **datagrams** internally, 数据报
 - this decision must be made anew for **every** arriving data **packet** since the best route may have changed since last time.
 - ✦ If the network uses **virtual circuits** internally, 虚电路
 - routing decisions are made only when a **new virtual circuit** is being set up.
 - is sometimes called **session routing**. 会话路由

- a router having two processes
 - ✦ One handles each packet as it arrives, **looking up** the outgoing line to use for it in the **routing tables**. 查找路由表
 - This process is **forwarding**.
 - ✦ The other process is responsible for **filling in** and **updating** the **routing tables**. 更新路由表
- certain properties are desirable in a routing algorithm
 - ✦ Correctness 正确性
 - ✦ Simplicity 简单性
 - ✦ Robustness 健壮性, 鲁棒性
 - ✦ Stability 稳定性
 - ✦ Fairness 公平性
 - ✦ and Efficiency 有效性

conflict between fairness and efficiency

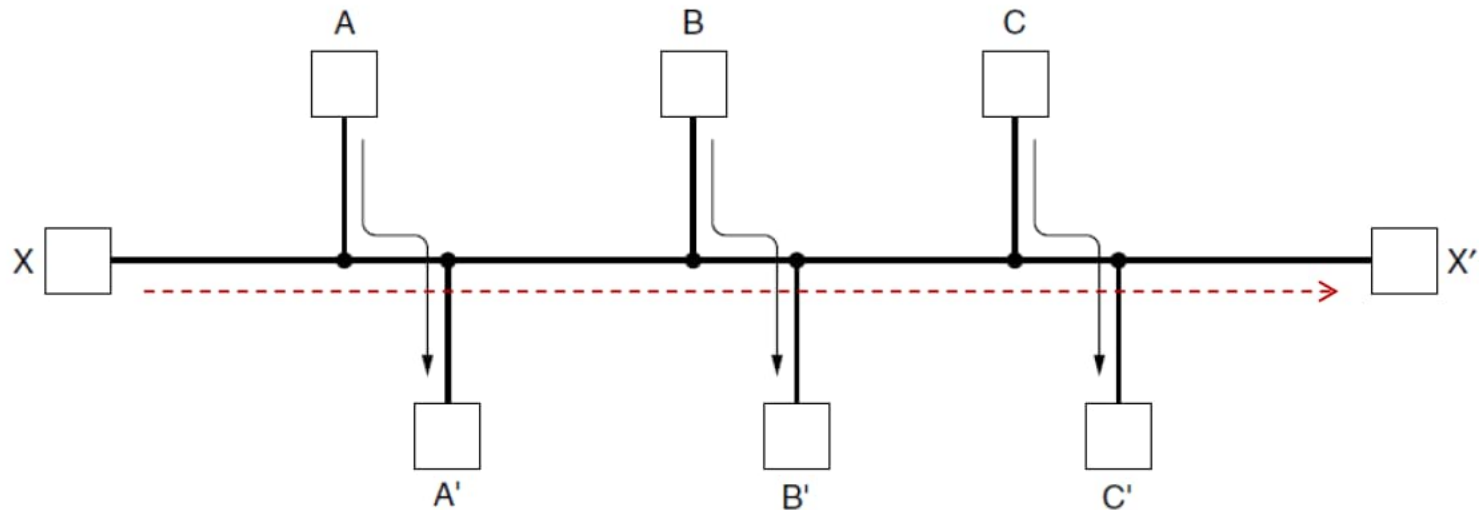


Figure 5-5. Network with a conflict between fairness and efficiency.

classes of routing algorithms



➤ Nonadaptive algorithms 非自适应算法

- ✦ the choice of the route to use is computed in advance, offline, and downloaded to the routers when the network is booted.
- ✦ is sometimes called static routing. 静态路由

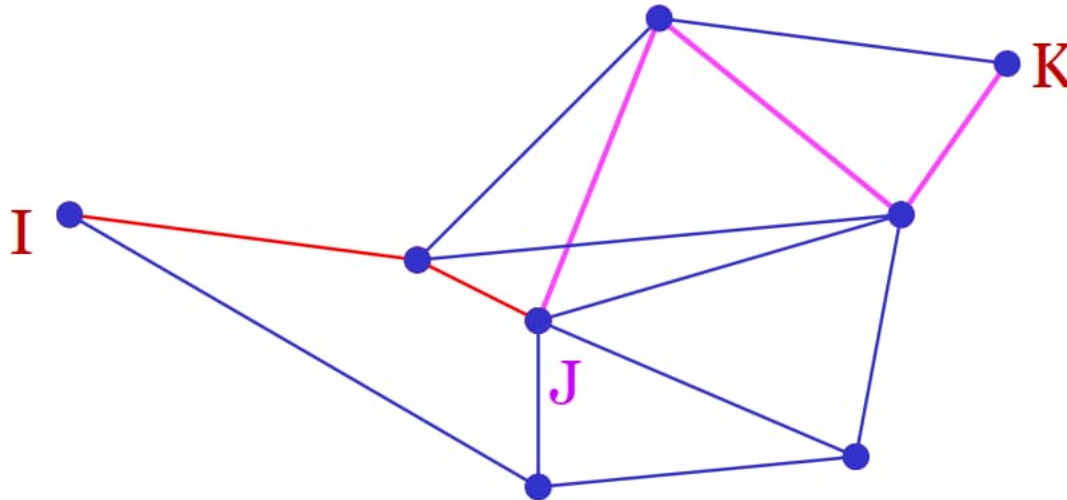
➤ Adaptive algorithms 自适应算法

- ✦ change their routing decisions to reflect changes in the topology, and sometimes changes in the traffic as well.
- ✦ dynamic routing algorithms 动态路由

5.2.1 The Optimality Principle

➤ The Optimality Principle 优化原理

- ✦ if router **J** is on the optimal path from router **I** to router **K**, then the optimal path from **J** to **K** also falls along the same route.



sink tree

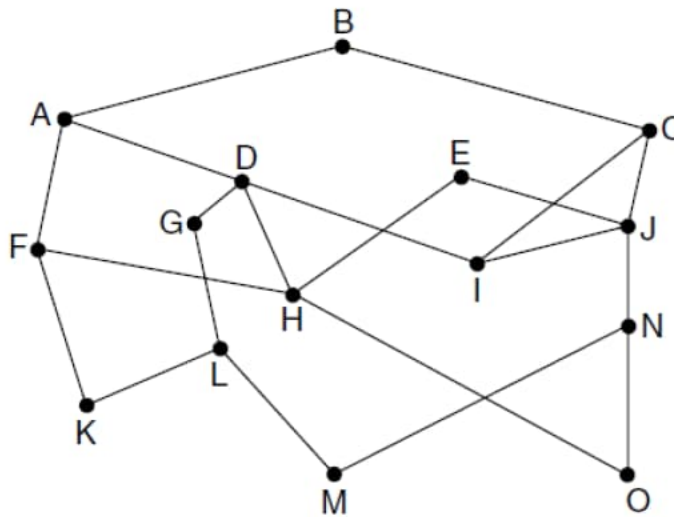
➤ sink tree 汇集树

✦ the set of optimal routes from all sources to a given destination form a tree rooted at the destination.

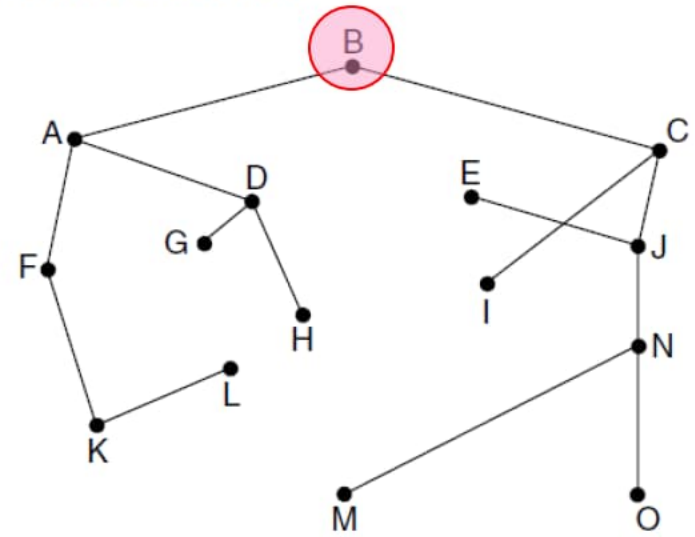
✦ DAG (Directed Acyclic Graph) 有向无环图

➤ The goal of all routing algorithms

✦ is to discover and use the sink trees for all routers.



(a) A network.



(b) A sink tree for router B.

Figure 5-6.

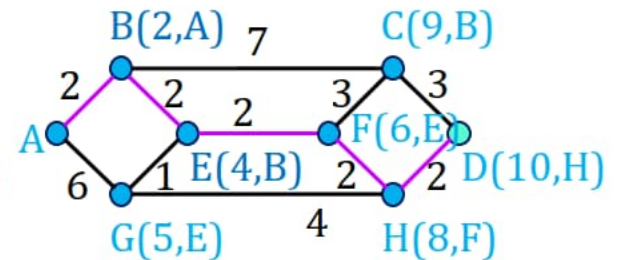
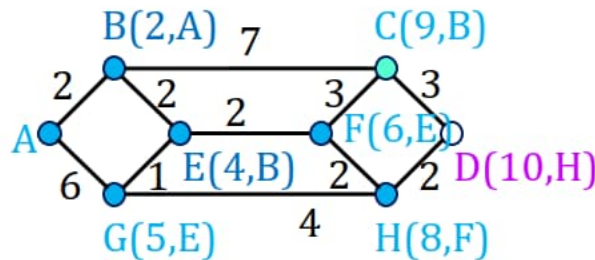
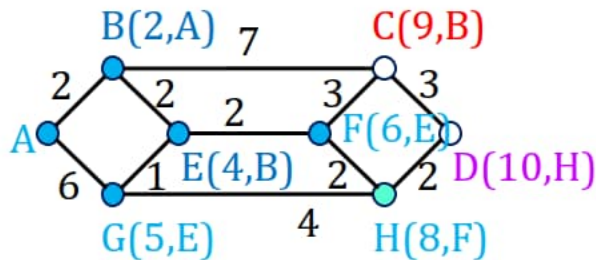
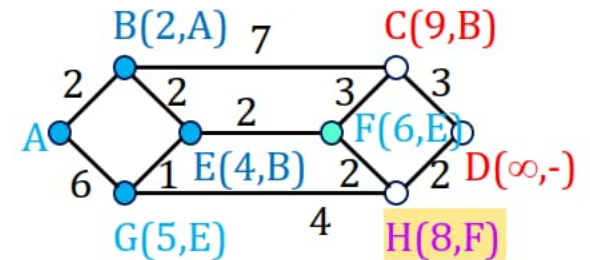
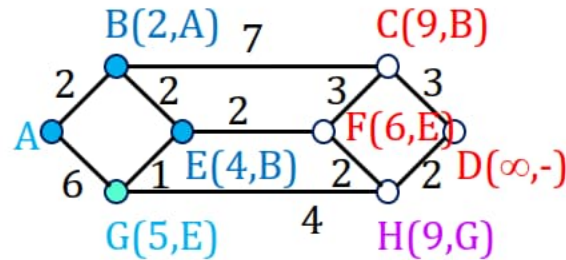
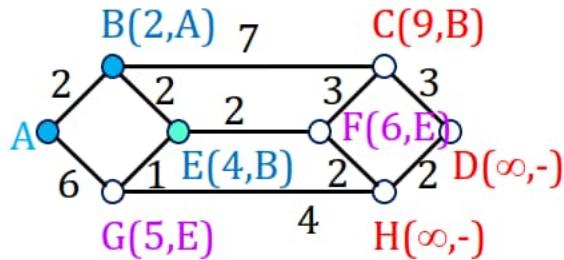
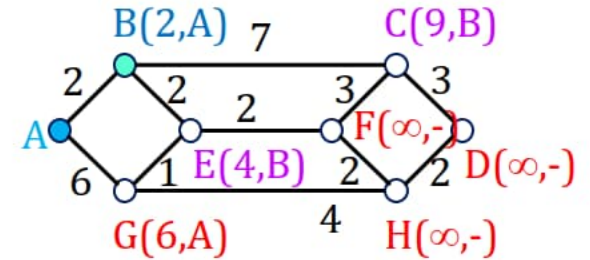
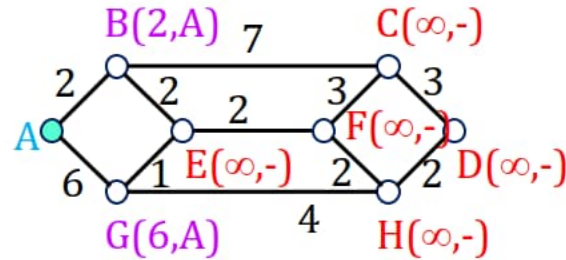
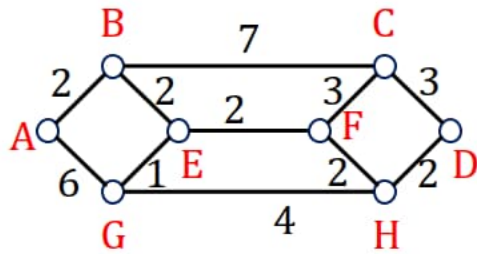
5.2.2 Shortest Path Algorithm

➤ shortest path

- ✦ One way of measuring path length is **the number of hops**. 跳数
- ✦ Another metric is **the geographic distance** in kilometers. 地理距离
- ✦ **delay** of a standard test packet. 延迟
- ✦ the labels on the edges could be computed **as a function** of 函数
 - the **distance**, **bandwidth**, **average traffic**, communication **cost**, measured **delay**, and **other factors**.
 - 距离, 带宽, 平均业务量, 通信成本, 测量延迟, 等等。

Dijkstra's algorithm

● working node ● permanent node



The shortest path is A-B-E-F-H-D, with distance 10.

5.2.3 Flooding

- Flooding 洪泛、泛洪
 - ✦ every incoming packet is sent out on every outgoing line except the one it arrived on.
- Classification: static routing
- problem
 - ✦ Flooding obviously generates vast numbers of duplicate packets.
- some measures to the problem
 - ✦ have a hop counter contained in the header of each packet that is decremented at each hop. (discarded when reaches zero)
 - ✦ have routers keep track of which packets have been flooded, to avoid sending them out a second time.
- uses
 - ✦ it ensures that a packet is delivered to every node in the network.
 - ✦ flooding is tremendously robust.
 - ✦ Flooding can also be used as a metric against which other routing algorithms can be compared.

measure ['meʒər] n. 措施; 衡量

5.2.4 Distance Vector Routing

- the most popular dynamic algorithms
 - ✦ DV: Distance Vector routing 距离矢量（向量）路由
 - ✦ LS: Link State routing 链路状态路由
- distance vector routing
 - ✦ each router maintain a table (i.e., a vector) giving the best known distance to each destination and which link to use to get there.
 - ✦ These tables are updated by exchanging information with the neighbors.
- entry in a routing table has two parts
 - ✦ the preferred outgoing line to use for that destination
 - ✦ and an estimate of the distance to that destination.

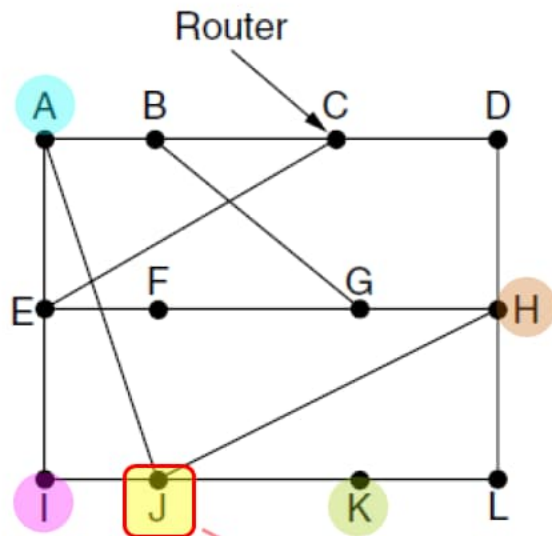


Figure 5-9. (a) A network.

To	A	I	H	K	New estimated delay from J	
A	0	24	20	21	8	A
B	12	36	31	28	20	A
C	25	18	19	36	28	I
D	40	27	8	24	20	H
E	14	7	30	22	17	I
F	23	20	19	40	30	I
G	18	31	6	31	18	H
H	17	20	0	19	12	H
I	21	0	14	22	10	I
J	9	11	7	10	0	—
K	24	22	22	0	6	K
L	29	33	9	9	15	K

JA delay is 8	JI delay is 10	JH delay is 12	JK delay is 6
---------------	----------------	----------------	---------------

Vectors received from J's four neighbors

New routing table for J	
8	A
20	A
28	I
20	H
17	I
30	I
18	H
12	H
10	I
0	—
6	K
15	K

(b) Input from A, I, H, K, and the new routing table for J.

The Count-to-Infinity Problem

➤ convergence

✦ The settling of routes to **best paths** across the network.

➤ DV has a serious drawback in practice

✦ it reacts **rapidly** to **good news**, but **leisurely** to **bad news**.

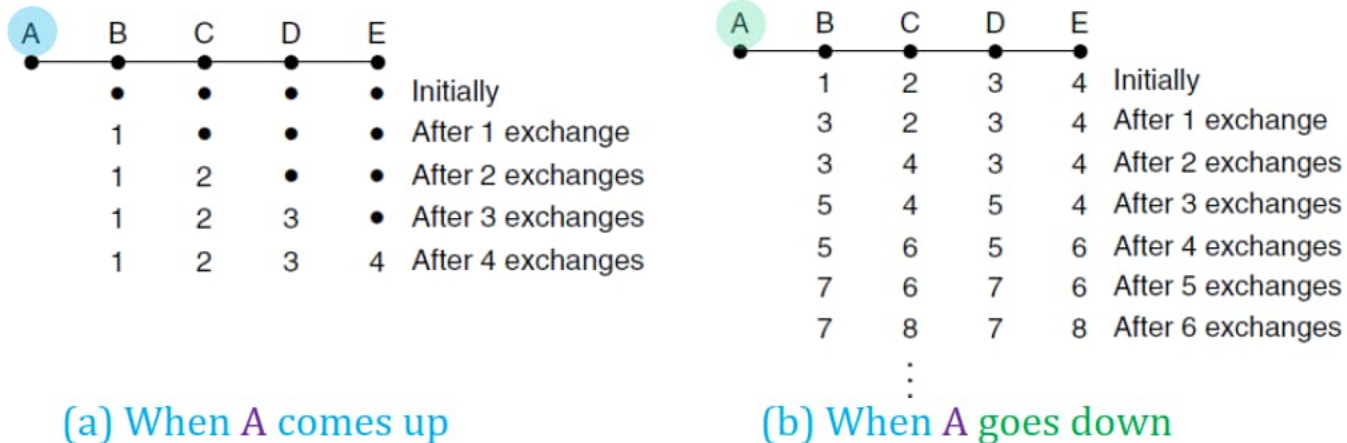


Figure 5-10. The count-to-infinity problem.

➤ RFC 1058

✦ the split horizon with poisoned reverse rule 抑制反转的视野分割

5.2.5 Link State Routing



- classification

- ✦ dynamic routing

- protocols

- ✦ IS-IS, OSPF

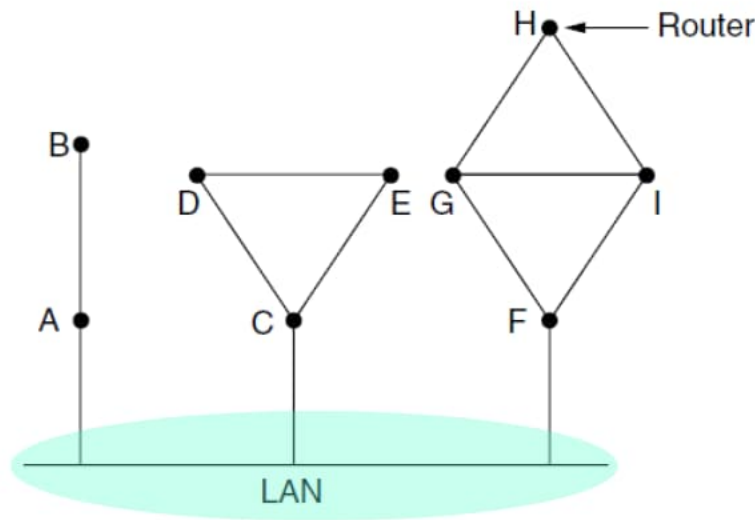
- five parts

- ✦ 1. Discover its neighbors and learn their network addresses.
- ✦ 2. Set the distance or cost metric to each of its neighbors.
- ✦ 3. Construct a packet telling all it has just learned.
- ✦ 4. Send this packet to and receive packets from all other routers.
- ✦ 5. Compute the shortest path to every other router.
 - Dijkstra's algorithm

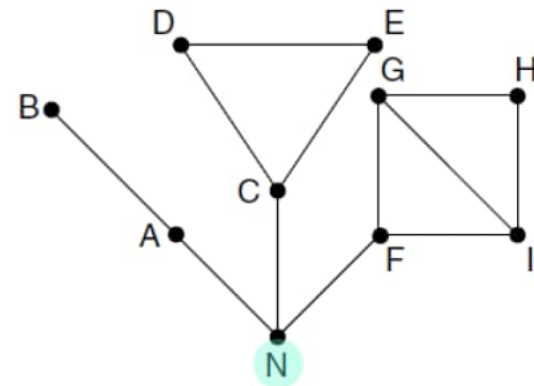
Learning about the Neighbors

➤ Learning 发现邻居

- ★ sending a special **HELLO packet** on each point-to-point line.



(a) Nine routers and a broadcast LAN.



DR: Designated Router
指派(虚拟的)路由器

(b) A graph model of (a).

Figure 5-11.

Setting Link Costs



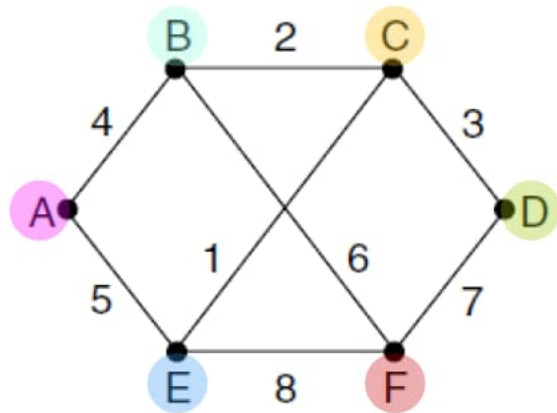
➤ Static 静态配置

- ✦ configured by the network operator

➤ Dynamic 动态设定

- ✦ the cost **inversely proportional** to the **bandwidth** of the link.
 - This makes **higher-capacity** paths better choices.
- ✦ the **delay** of the links may be factored into the cost.
 - paths over **shorter links** are better choices.
 - The most direct way to **determine** this **delay** is to send over the line a special **ECHO packet** that the other side is required to send back immediately.

Building Link State Packets



(a) A network.

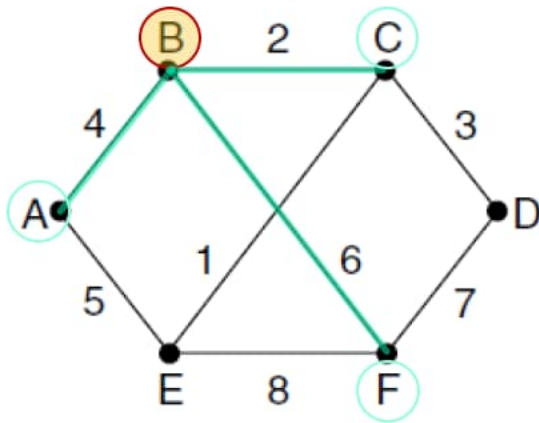
Link State Packets

A	B	C	D	E	F
Seq.	Seq.	Seq.	Seq.	Seq.	Seq.
Age	Age	Age	Age	Age	Age
B 4	A 4	B 2	C 3	A 5	B 6
E 5	C 2	D 3	F 7	C 1	D 7
	F 6	E 1		F 8	E 8

(b) The link state packets for this network.

Figure 5-12

Distributing the Link State Packets



Source	Seq.	Age	Send flags			ACK flags			Data
			A	C	F	A	C	F	
A	21	60	0	1	1	1	0	0	
F	21	60	1	1	0	0	0	1	
E	21	59	0	1	0	1	0	1	
C	20	60	1	0	1	0	1	0	
D	21	59	1	0	0	0	1	1	

Figure 5-13. The packet buffer for router B

Computing the New Routes



- ✦ construct the entire network graph 构建全网拓扑图
- ✦ Every link is represented **twice**, once for **each direction**. 双向成本
 - The **different directions** may even have **different costs**.
- ✦ **Dijkstra's algorithm** can be run **locally**. 本地运行
- ✦ This information is installed in the **routing tables**. 生成路由表

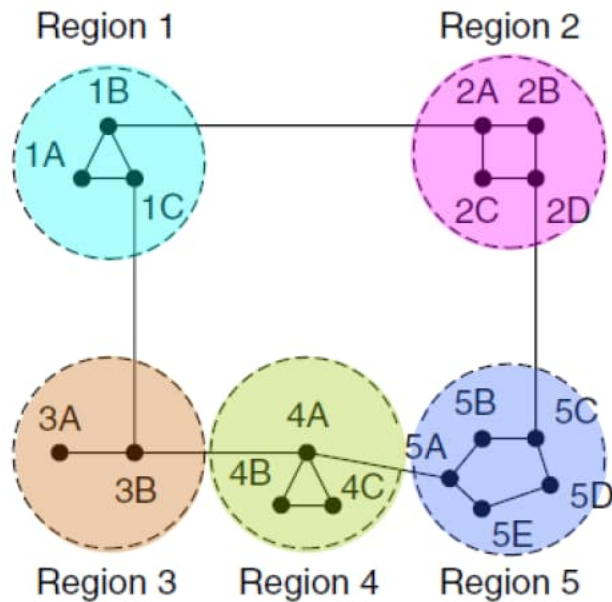
Comparison to DV



- Compared to distance vector routing, link state routing
 - ✦ requires more memory and computation
 - ✦ works well because it does not suffer from slow convergence problems
 - ✦ is widely used in actual networks
 - IS-IS (Intermediate System-Intermediate System) 中间系统到中间系统
 - OSPF (Open Shortest Path First) 开放最短路径优先

- Packets transmission
 - ✦ DV: delivering routing table about network to neighbors
 - 向邻居发送全网信息
 - ✦ LS: delivering neighbors' information to all routers in the network
 - 向全网发送邻居信息

5.2.6 Hierarchical Routing



Full table for 1A

Dest.	Line	Hops
1A	—	—
1B	1B	1
1C	1C	1
2A	1B	2
2B	1B	3
2C	1B	3
2D	1B	4
3A	1C	3
3B	1C	2
4A	1C	3
4B	1C	4
4C	1C	4
5A	1C	4
5B	1C	5
5C	1B	5
5D	1C	6
5E	1C	5

Hierarchical table for 1A

Dest.	Line	Hops
1A	—	—
1B	1B	1
1C	1C	1
2	1B	2
3	1C	2
4	1C	3
5	1C	4

Figure 5-14. Hierarchical routing.

5.2.7 Broadcast Routing-1



➤ broadcast 广播

- ✦ Sending a packet to **all** destinations simultaneously

➤ broadcasting methods

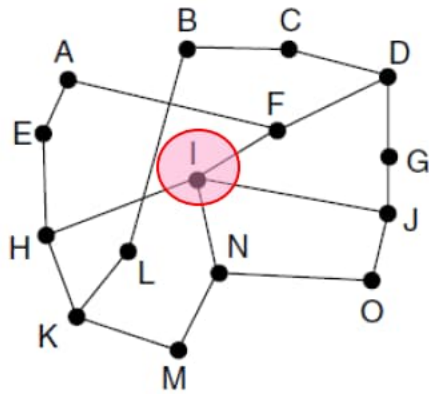
- ✦ send a **distinct** packet to each destination 单独发送
 - **wasteful** of bandwidth and **slow**
 - requires the source to have a **complete** list of **all** destinations
- ✦ multdestination routing 多目标路由
 - each packet contains either **a list** of destinations or **a bit map** indicating the desired destinations
 - requires the **source to know all** the destinations
 - a **router** must **to determine where to send** one multdestination packet as it is for multiple distinct packets
- ✦ flooding 洪泛

Broadcast Routing-2

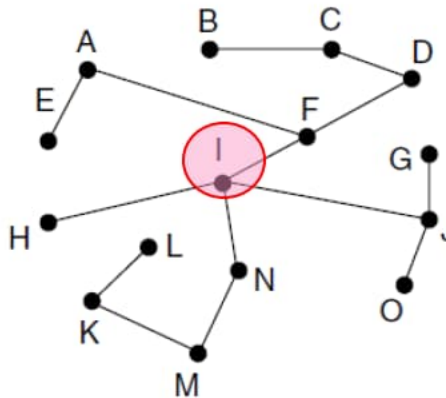


★ Reverse path forwarding 逆向路径转发

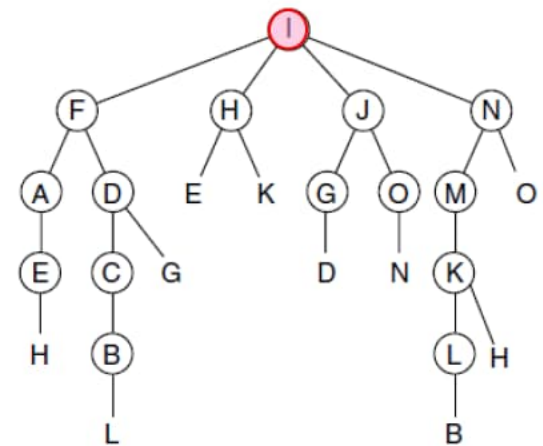
- the router **forwards** copies of it onto all links **except** the one it **arrived on**.
- Sink trees (汇集树), or spanning trees (生成树), **built by** reverse path forwarding



(a) A network.



(b) A sink tree.



(c) The **tree** built by reverse path forwarding.

Figure 5-15. Reverse path forwarding.

5.2.8 Multicast Routing

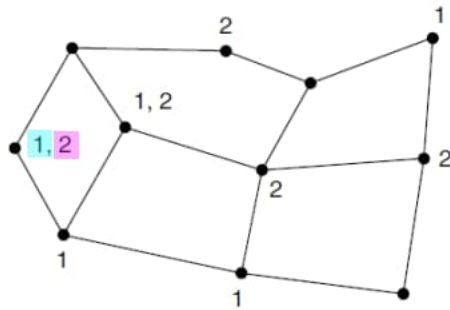


- multicast 组播/多播
 - ✦ to send messages to well-defined groups
- multicast routing
 - ✦ the routing algorithm used for multicast
- assume
 - ✦ that each group is identified by a multicast address and that routers know the groups to which they belong
- routing schemes
 - ✦ sending packets along spanning trees to deliver the packets to the members of the group

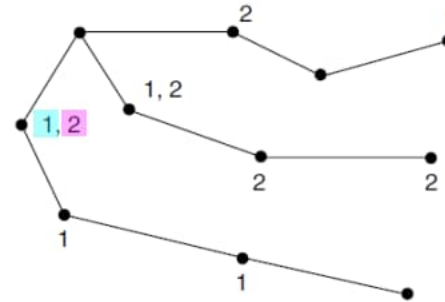
prune the broadcast spanning tree



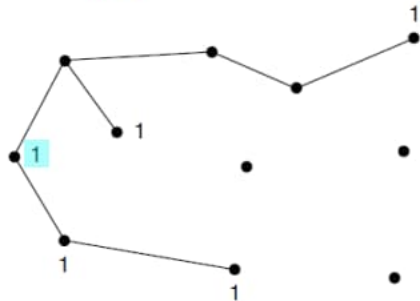
广播生成树剪枝



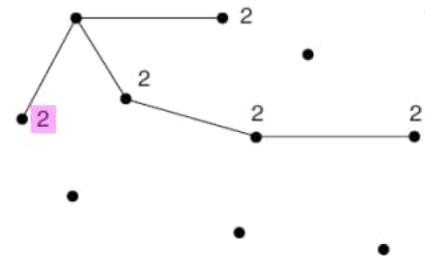
(a) A network.



(b) A spanning tree for the leftmost router.



(c) A multicast tree for group 1.



(d) A multicast tree for group 2.

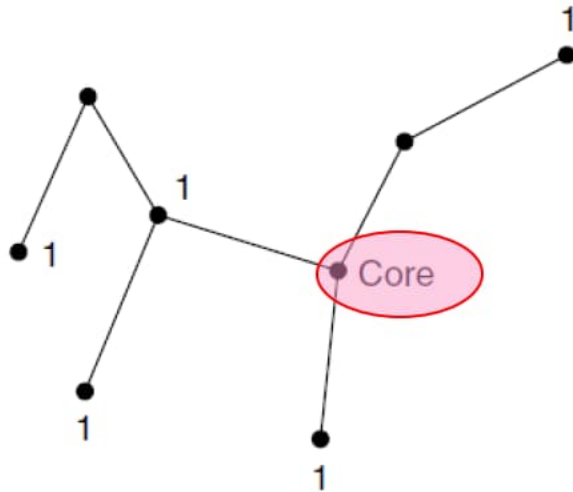
Figure 5-16.

- ✦ MOSPF: Multicast OSPF
- ✦ DVMRP: Distance Vector Multicast Routing Protocol

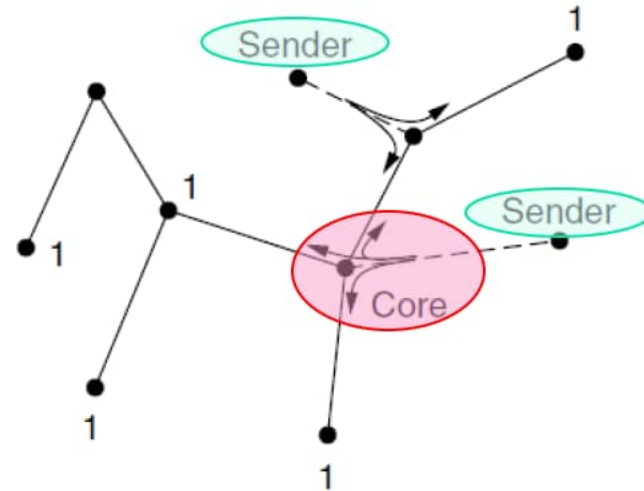
core-based trees



基于核心节点的组播树



(a) Core-based tree for group 1.



(b) Sending to group 1.

Figure 5-17.

★ PIM: Protocol Independent Multicast 协议无关性组播

5.2.9 Anycast Routing

- anycast 任播, 泛播, 选播
 - ✦ a packet is delivered to the nearest member of a group
- anycast routing
 - ✦ Schemes that find these paths

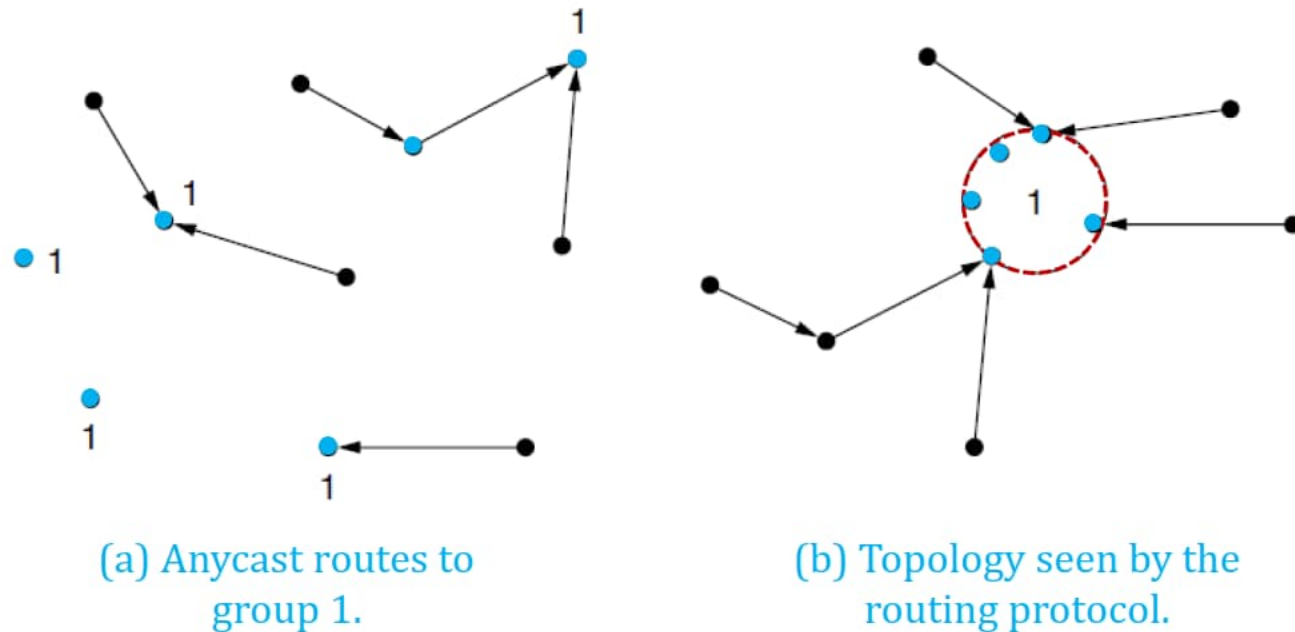


Figure 5-18.

5.2.10 Routing for Mobile Hosts

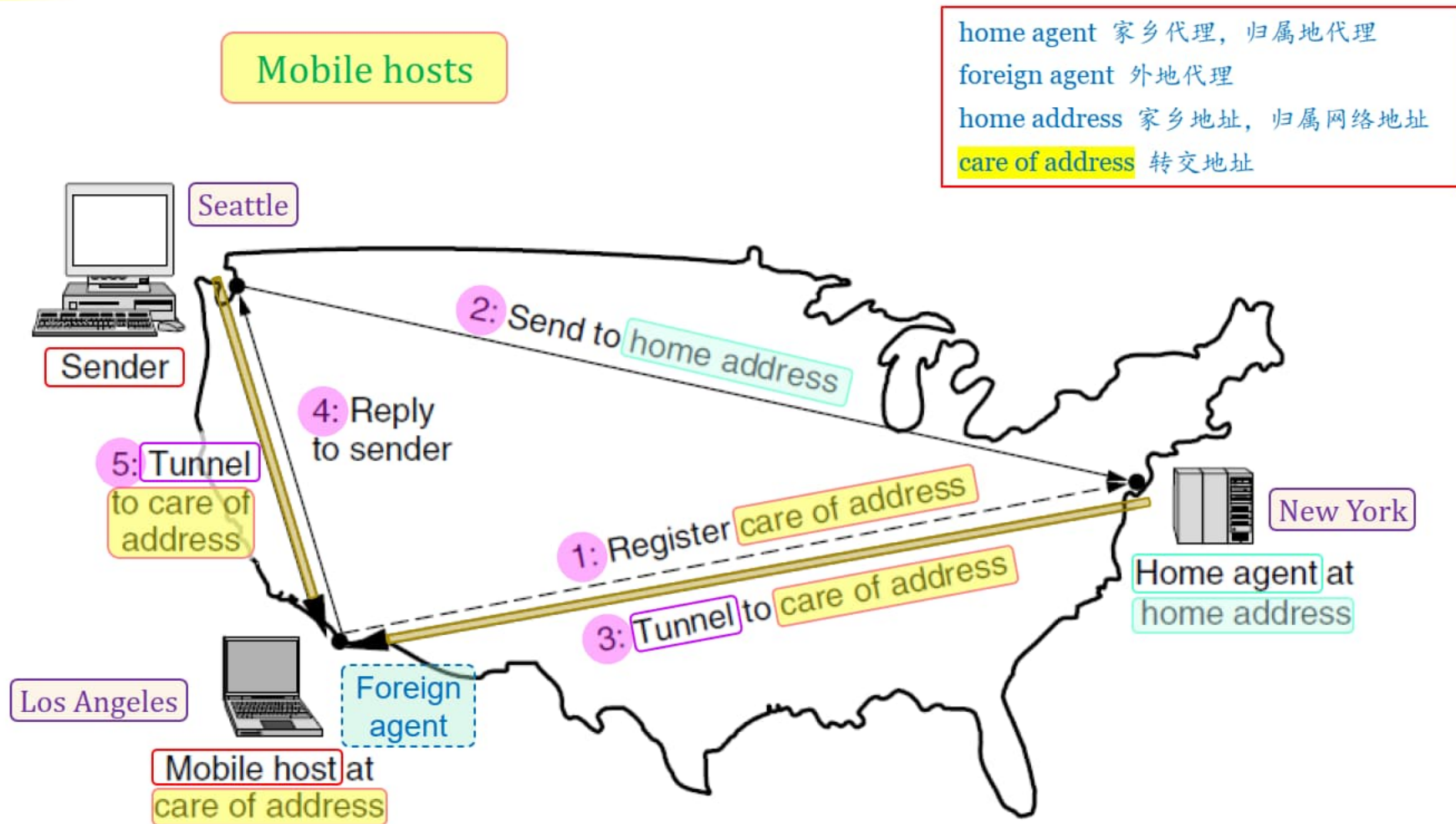


Figure 5-19. Packet routing for mobile hosts.

5.2.11 Routing in Ad Hoc Networks

➤ MANETs (Mobile Ad hoc NETWORKs) 移动自组织网络

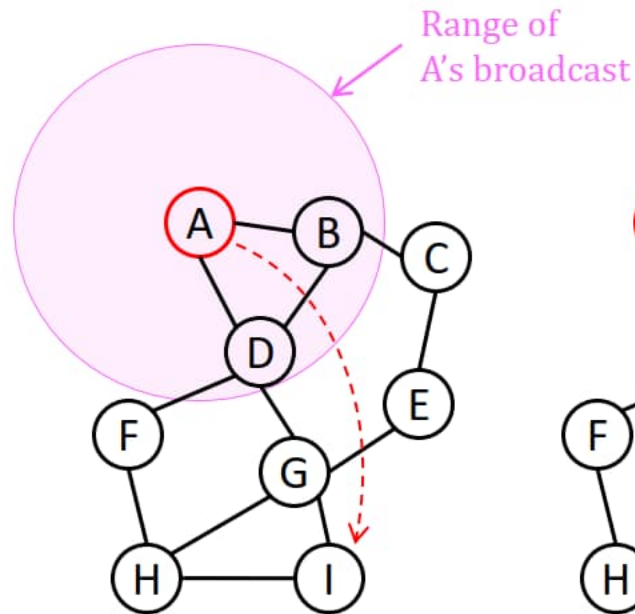
- ✦ the routers themselves are mobile

Mobile routers

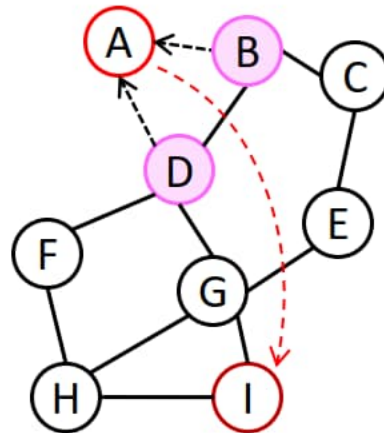
➤ AODV (Ad hoc On-demand Distance Vector) Ad hoc按需距离矢量

- ✦ routes to a destination are discovered on demand,
 - only when a somebody wants to send a packet to that destination.
- ✦ assumed model
 - each node can communicate with all other nodes that lie within its coverage circle.
 - all connections are symmetric.
- ✦ routing stages
 - Route Discovery
 - Route Maintenance

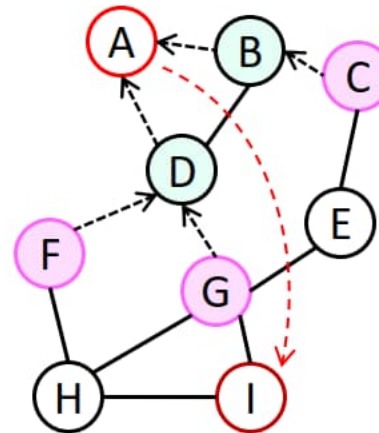
Route Discovery



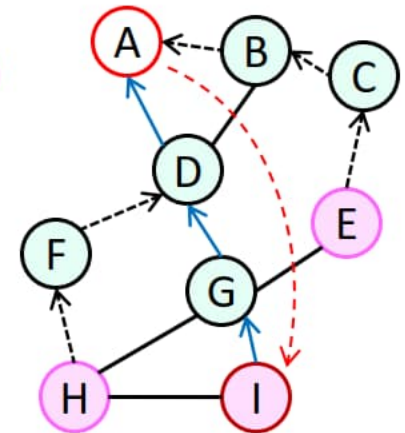
(a) Range of A's broadcast.



(b) After B and D receive it.



(c) After C, F, and G receive it.



(d) After E, H, and I receive it.

The pink shaded nodes are new recipients.
The dashed lines show possible reverse routes.
The solid blue lines show the discovered route.

Figure 5-20.

Route Maintenance



- ✦ Each node broadcasts a Hello message periodically
- ✦ Each of its neighbors is expected to respond to it.
 - If no response is forthcoming
 - the neighbor is no longer available, the invalid route is purged from the network.
 - senders can find new, valid routes by using the discovery mechanism that we described.



5.3 Congestion Control Algorithms



➤ Congestion 拥塞

- ✦ Too many packets present in (a part of) the network causes **packet delay and loss** that degrades performance.

➤ Congestion control 拥塞控制

- ✦ requires the **network** and **transport layers** to work together.

➤ the onset of congestion 拥塞的产生

- ✦ **bursts of traffic** occasionally fill up the buffers inside routers
 - and some packets are **lost**.
- ✦ These **lost** packets **consume** some of the **capacity**,
 - so the number of delivered packets falls below the ideal curve.
- ✦ The network is now **congested**.

onset 发作

➤ congestion collapse 拥塞崩溃

- ★ performance **plummets** as the offered load increases beyond the capacity.

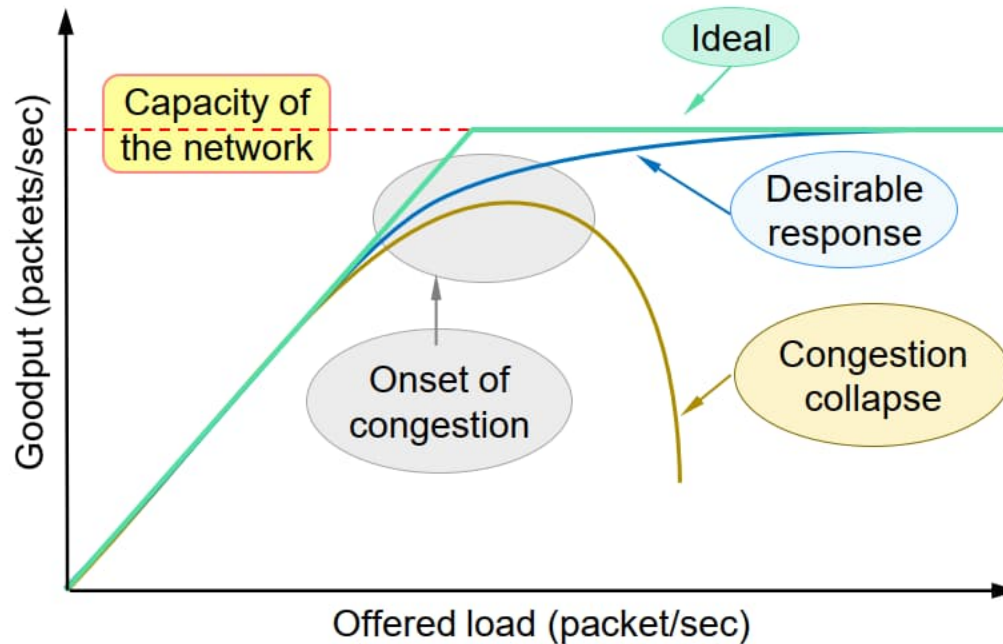


Figure 5-21. With too much traffic, performance **drops sharply**.

plummet 暴跌, 骤降, 断崖式下跌
goodput 有效吞吐量

causes



➤ causes

- ✦ when packets are greatly delayed
- ✦ senders retransmit them, thinking that they have been lost
- ✦ copies of the same packet will be delivered by the network,
 - again wasting its capacity

- Nagle (1987) realized that
 - ✦ if routers have an infinite amount of memory, congestion gets worse, not better.
- This is because
 - ✦ by the time packets get to the front of the queue,
 - ✦ they have already timed out (repeatedly)
 - ✦ and duplicates have been sent.
 - ✦ This makes matters worse, not better
 - ✦ it leads to congestion collapse.

Congestion control and Flow control



- the difference between congestion control and flow control
 - ✦ Congestion control has to do with making sure the network is able to carry the offered traffic.
 - It is a global issue, involving the behavior of all the hosts and routers.
 - ✦ Flow control, in contrast, relates to the traffic between a particular sender and a particular receiver.
 - Its job is to make sure that a fast sender cannot continually transmit data faster than the receiver is able to absorb it.
- handle both problems
 - ✦ get the host to slow down.

5.3.1 Approaches to Congestion Control



➤ Cause:

- ✦ the **load** is (temporarily) **greater than** the **resources** (in a part of the network) can handle

➤ Two solutions to congestion

- ✦ **increase** the **resources**
- ✦ or **decrease** the **load**.



Figure 5-22. Timescales of approaches to congestion control.

Congestion avoidance

- network provisioning 网络供给
 - ✦ add resources dynamically when there is serious congestion
- traffic-aware routing 业务量感知路由
 - ✦ Splitting traffic across multiple paths
- admission control 准入控制
 - ✦ new connections can be refused if they would cause the network to become congested
- Traffic Throttling 业务量限制
 - ✦ congestion avoidance
- load shedding 负载脱落
 - ✦ to discard packets that it cannot deliver

throttle *v.* 压制, 限制, 阻挡
shed *v.* 脱落, 抛弃

5.3.2 Traffic-Aware Routing



业务量感知路由

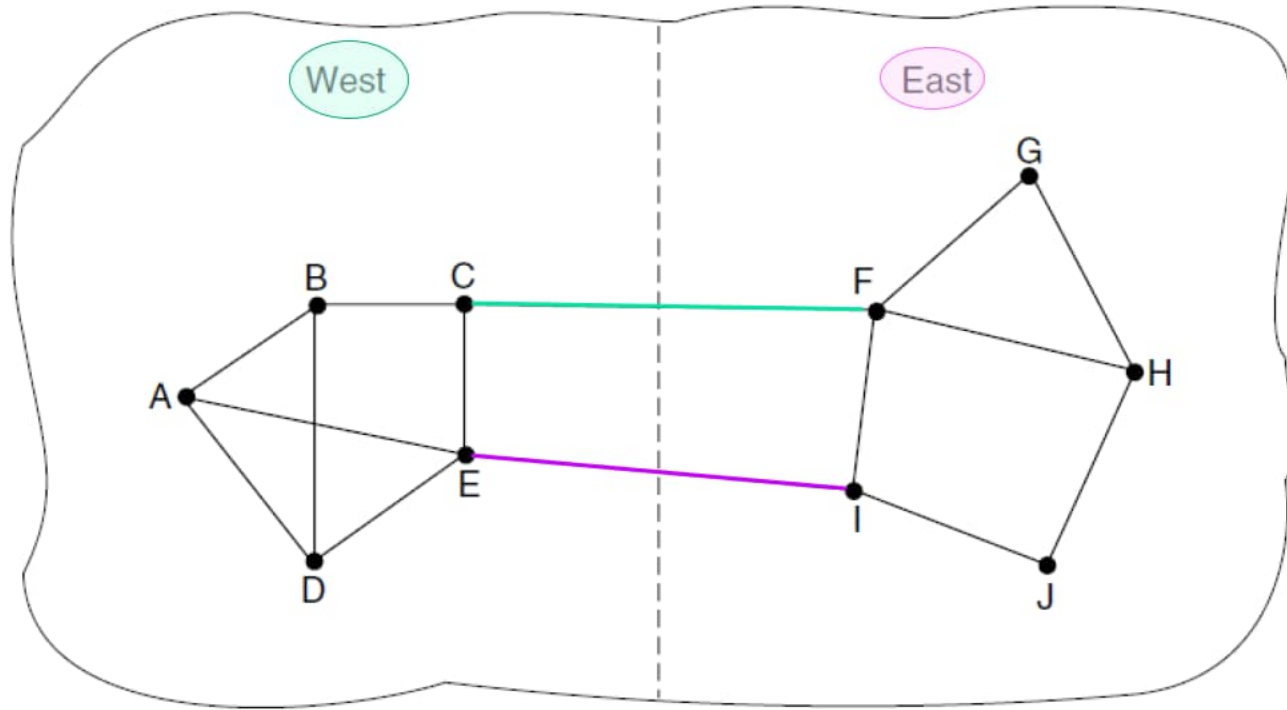


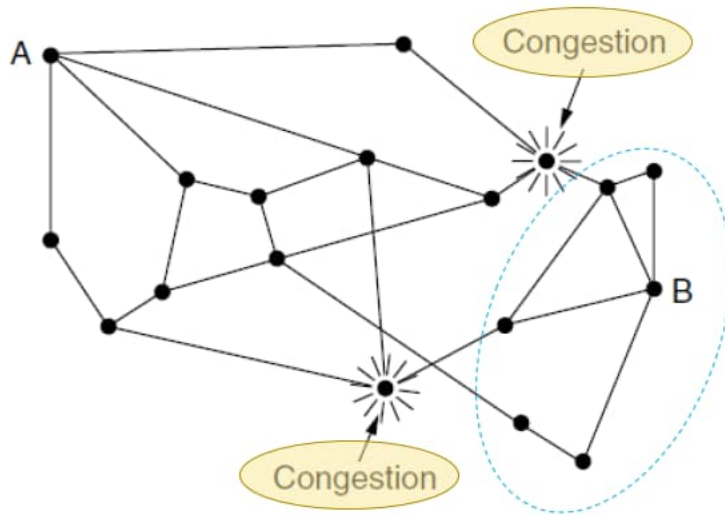
Figure 5-23. A network in which the East and West parts are connected by two links.

➤ traffic engineering 业务量工程

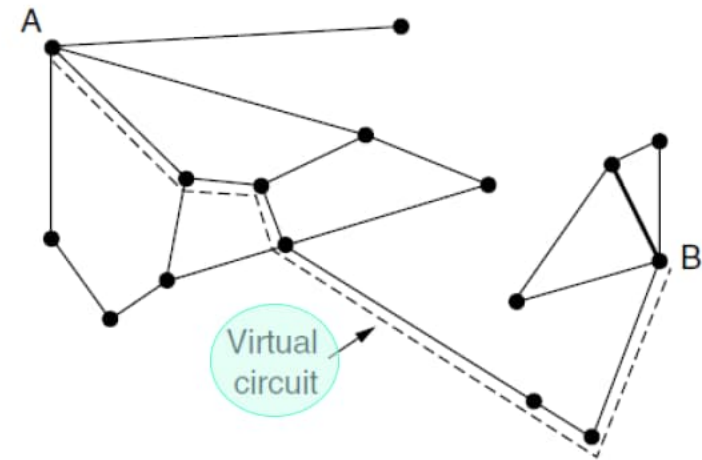
5.3.3 Admission Control



准入控制



(a) A congested network.



(b) The portion of the network that is not congested.

Figure 5-24. A virtual circuit from A to B is also shown.

5.3.4 Traffic Throttling

业务量限制

➤ congestion avoidance 拥塞避免

- ✦ routers must **determine** when **congestion is approaching**, before it has arrived
 - the **utilization** of the output links 输出链路利用率
 - the **buffering** of queued packets inside the router 队列长度
 - The **queueing delay** inside routers 排队延迟
 - **EWMA**: **E**xponentially **W**eighted **M**oving **A**verage 指数加权移动平均
 - $d_{new} = \alpha \cdot d_{old} + (1 - \alpha) \cdot s$
 - the number of **packets** that are **lost** due to insufficient buffering
- ✦ routers must **deliver timely feedback to the senders** that are causing the congestion.

➤ Choke Packets 分组抑制

- ✦ **SOURCE QUENCH** message 信源抑制报文

choke *v. n.* 阻塞, 抑制
quench *vt.* 熄灭, 扑灭

Explicit Congestion Notification



➤ ECN: Explicit Congestion Notification 显式拥塞通知

★ RFC 3168, The Addition of Explicit Congestion Notification (ECN) to IP

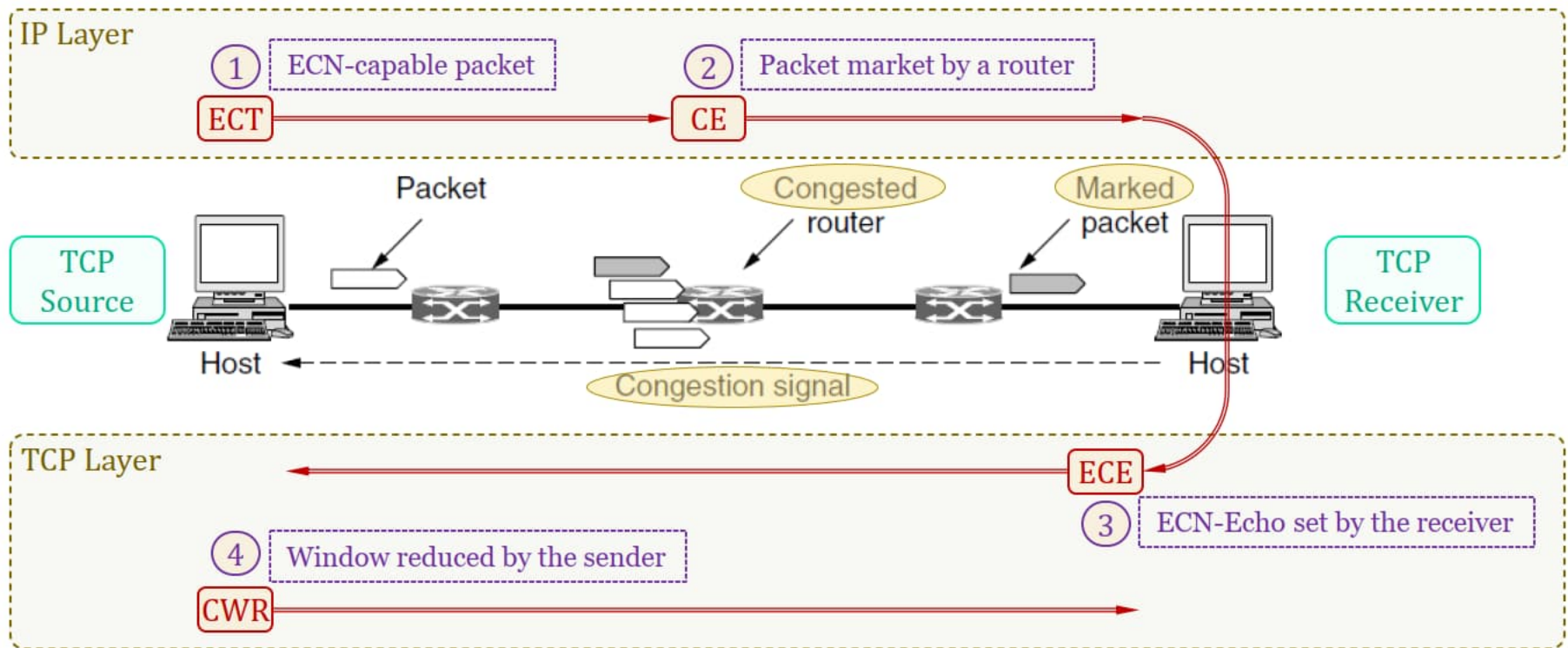


Figure 5-25. Explicit congestion notification

RFC 3168 The Addition of Explicit Congestion Notification (ECN) to IP

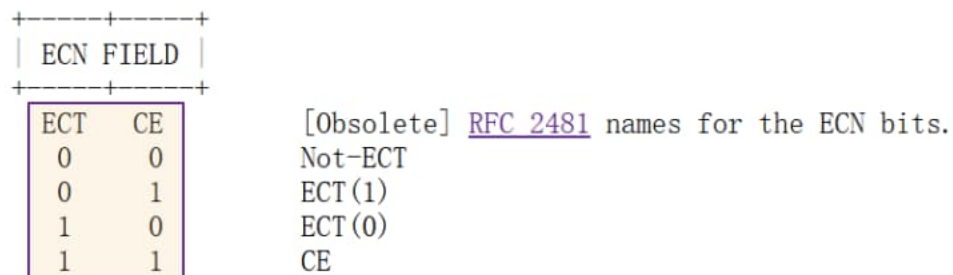
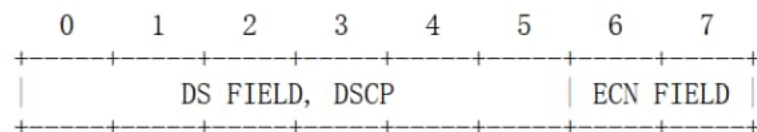


Figure 1: The ECN Field in IP.



DSCP: differentiated services codepoint

ECN: Explicit Congestion Notification

Figure 2: The Differentiated Services and ECN Fields in IP.

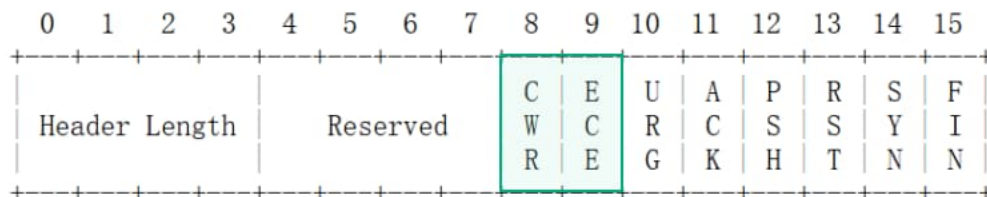
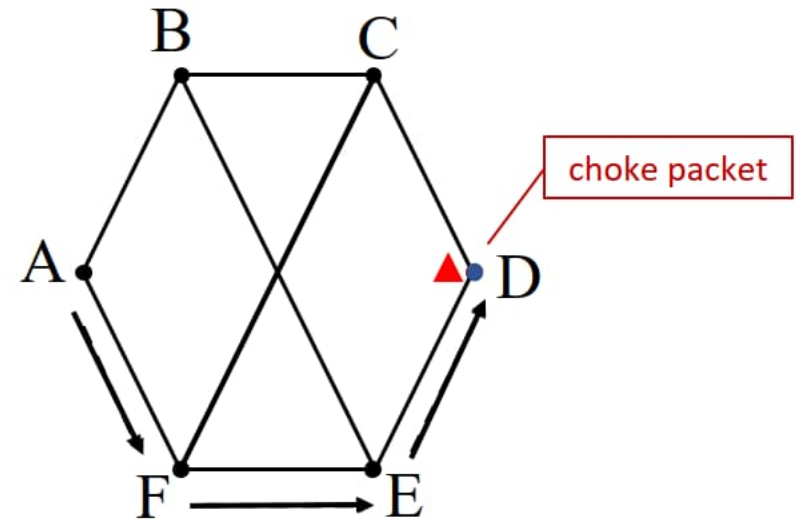
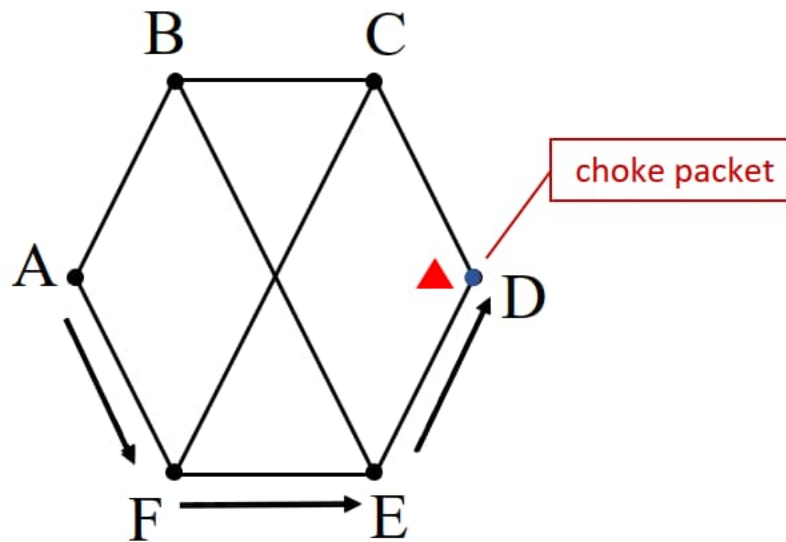


Figure 4: The new definition of bytes 13 and 14 of the TCP Header.

Hop-by-Hop Backpressure

➤ Hop-by-Hop Backpressure 逐跳后压



5.3.5 Load Shedding



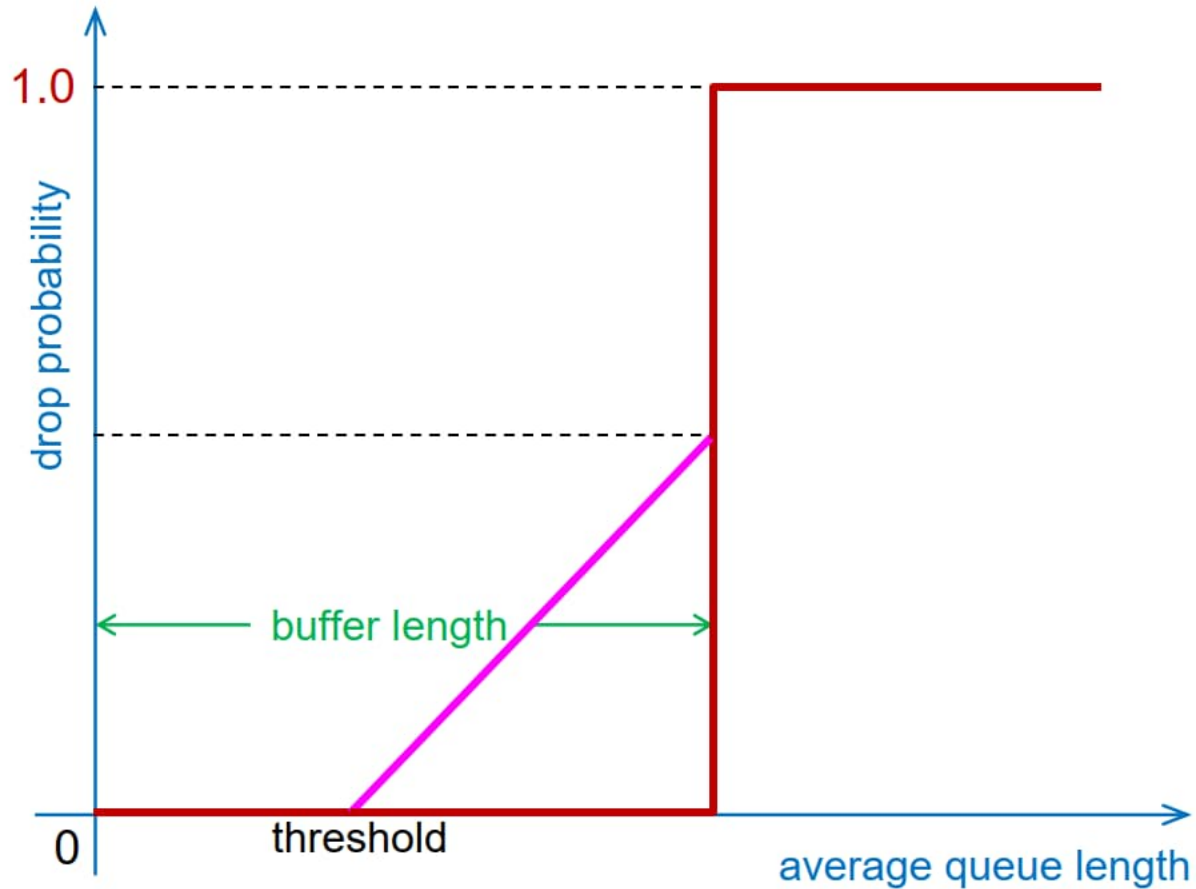
负载脱落

- which packets to drop,
 - ✦ depend on the **type** of applications
 - **wine**, old is better than new 葡萄酒策略
 - file transfer
 - **milk**, new is better than old 牛奶策略
 - real-time media

Random Early Detection



- RED: Random Early Detection 随机早期检测



5.4 Quality of Service

- QoS: Quality of Service 服务质量
- Overprovisioning 过度资源配置
 - ✦ build a network with enough capacity for whatever traffic will be thrown at it.
 - telephone system
 - ✦ The trouble with this solution
 - expensive
 - based on expected traffic
- Four issues must be addressed to ensure quality of service
 - ✦ What applications need from the network.
 - ✦ How to regulate the traffic that enters the network.
 - ✦ How to reserve resources at routers to guarantee performance.
 - ✦ Whether the network can safely accept more traffic.

5.4.1 Application Requirements



	带宽	延迟	抖动	丢失
Application	Bandwidth	Delay	Jitter	Loss
Email	Low	Low	Low	Medium
File sharing	High	Low	Low	Medium
Web access	Medium	Medium	Low	Medium
Remote login	Low	Medium	Medium	Medium
Audio on demand	Low	Low	High	Low
Video on demand	High	Low	High	Low
Telephony	Low	High	High	Low
Videoconferencing	High	High	High	Low

Figure 5-27. Stringency of applications' quality-of-service requirements.

➤ Technologies for QoS

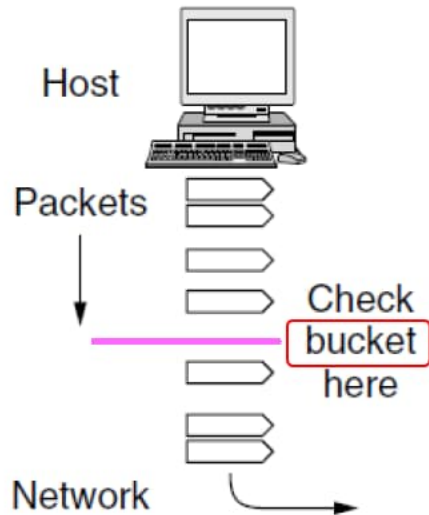
- ✦ Traffic Shaping 业务量整形
 - regulate the burstiness of a flow
- ✦ Packet Scheduling 分组调度
 - Reserve sufficient resources along the route
- ✦ Admission Control 准入控制
 - accept or reject the flow based on its policy
- ✦ Integrated Services 综合服务
 - RSVP for streaming multimedia
- ✦ Differentiated Services 区分服务
 - class-based QoS

5.4.2 Traffic Shaping

- bursty 突发性
 - ✦ at nonuniform rates
- Traffic shaping 业务量整形
 - ✦ regulating the average rate and burstiness of a flow
- SLA (Service Level Agreement) 服务等级协定
 - ✦ the customer and the provider agree on a certain traffic pattern
- traffic policing 业务量监管
 - ✦ Monitoring a traffic flow

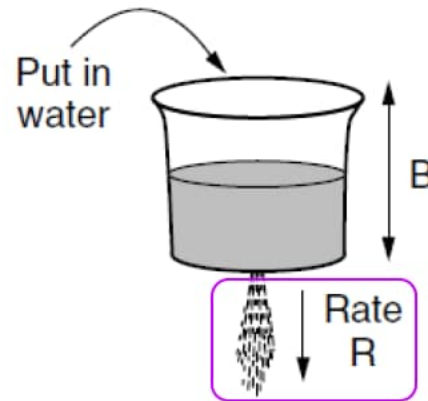
Leaky and Token Buckets

漏桶和令牌桶



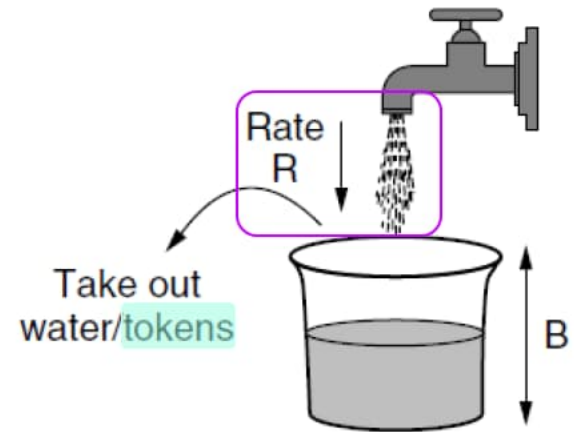
(a) Shaping packets.

分组整形



(b) A leaky bucket.

漏桶



(c) A token bucket.

令牌桶

Figure 5-28.

5.4.3 Packet Scheduling



- Packet scheduling 分组调度
 - ✦ Reserve sufficient resources along the route
 - ✦ Three different kinds of resources can potentially be reserved for different flows:
 - 1. Bandwidth. 带宽
 - 2. Buffer space. 缓存
 - 3. CPU cycles. CPU周期 (time slices)
- Packet scheduling algorithms 分组调度算法
 - ✦ allocate bandwidth and other router resources by determining which of the buffered packets to send on the output line next.
 - FIFO (First-In First-Out), FCFS (First-Come First-Serve)
 - fair queueing algorithm

Round-robin service



round robin 轮询

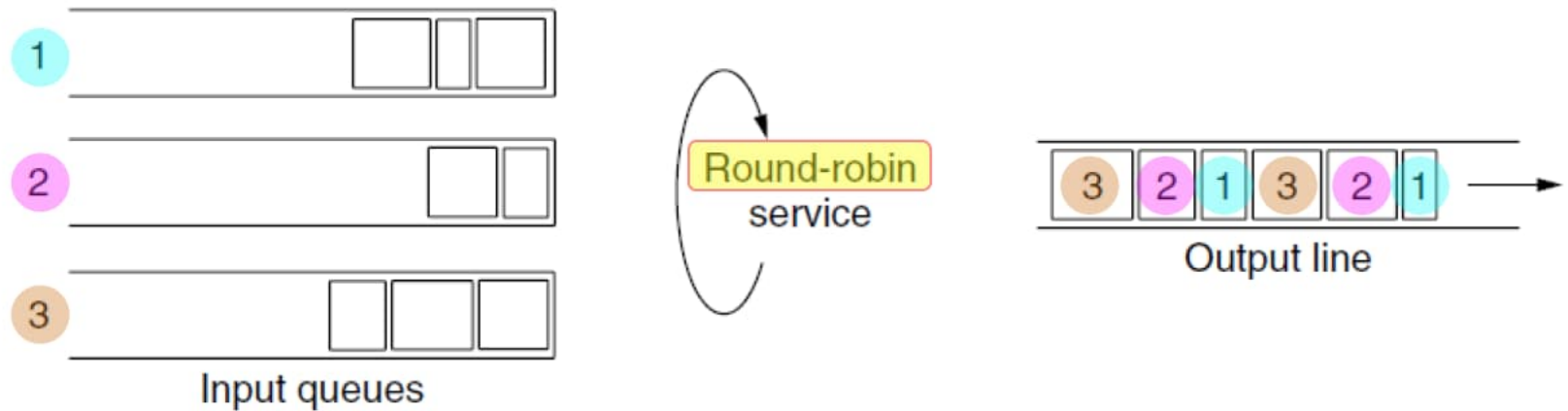
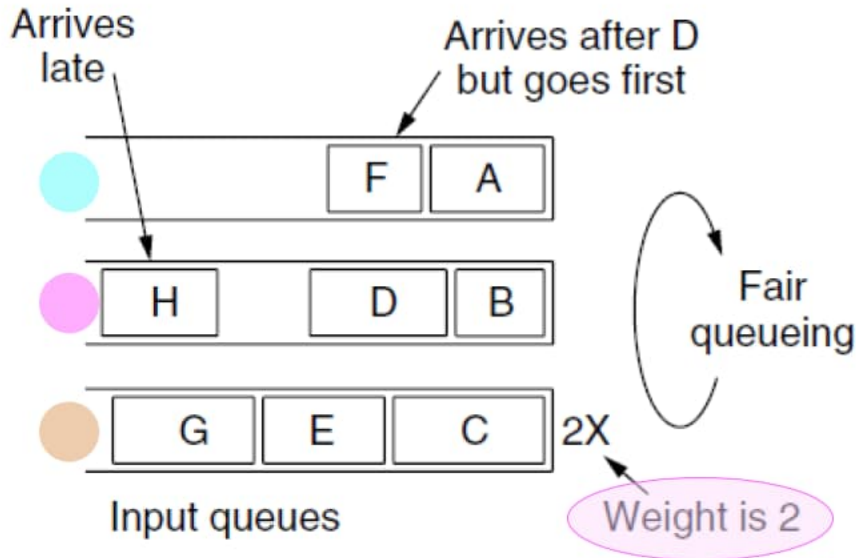


Figure 5-30. Round-robin fair queueing.

Weighted Fair Queueing



Packet	Arrival time	Length	Finish time	Output order
A	0	8	8	1
B	5	6	11	3
C	5	10	10	2
D	8	9	20	7
E	8	8	14	4
F	10	6	16	5
G	11	10	19	6
H	20	8	28	8

Output: A C B E F G D H

(a) Weighted Fair Queueing.

加权公平队列

(b) Finishing times for the packets.

Figure 5-31.

5.5 Internetworking

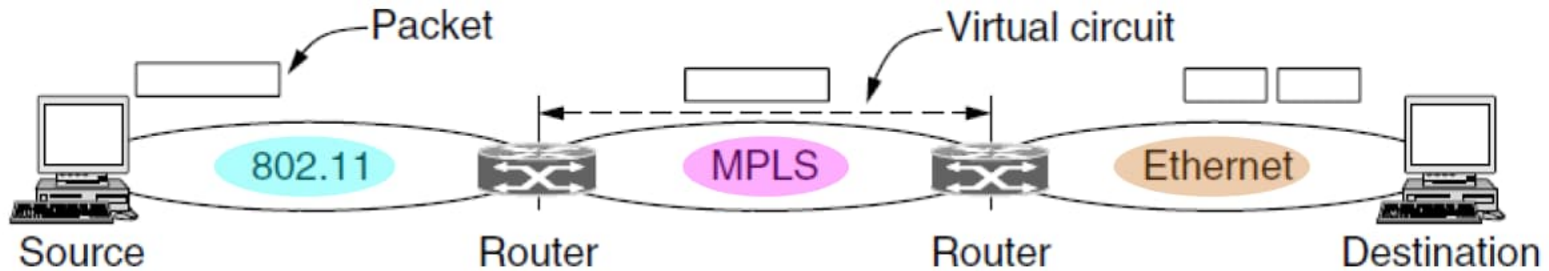
- internetwork, **internet** 网络互联, 互联网
 - ✦ **two** or **more networks** are connected
- The Internet 因特网
 - ✦ the prime **example** of this interconnection

5.5.1 How Networks Differ

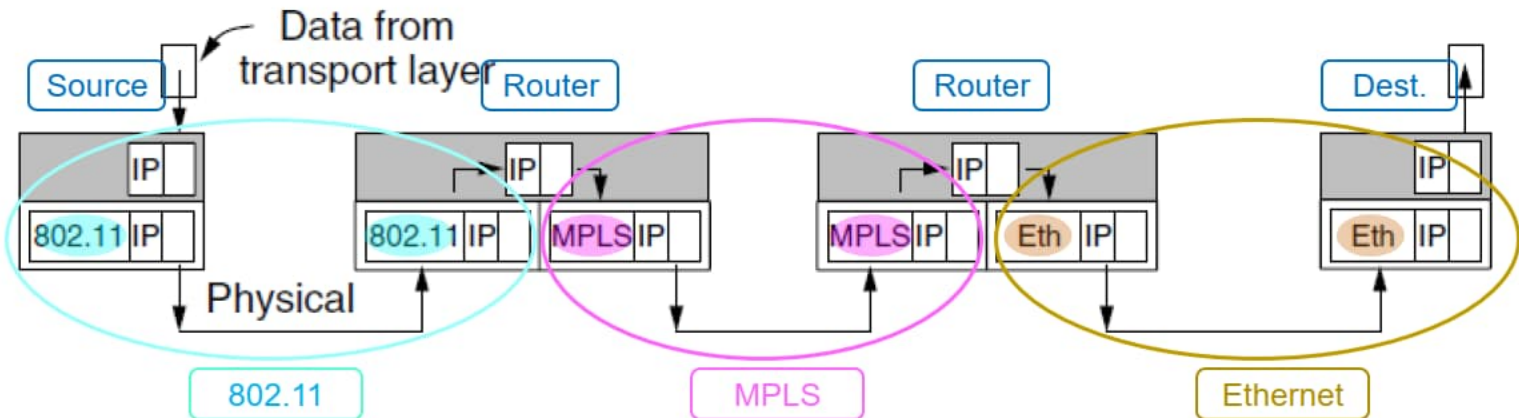
Item	Some Possibilities
Service offered	Connectionless versus connection oriented
Addressing 编址	Different sizes, flat or hierarchical
Broadcasting	Present or absent (also multicast)
Packet size	Every network has its own maximum
Ordering 按序	Ordered and unordered delivery
Quality of Service	Present or absent; many different kinds
Reliability	Different levels of loss
Security	Privacy rules, encryption, etc.
Parameters	Different timeouts, flow specifications, etc.
Accounting 计费	By connect time, packet, byte, or not at all

Figure 5-38. Some of the many ways networks can differ.

5.5.2 How Networks Can Be Connected



(a) A packet crossing different networks.



(b) Network and link layer protocol processing.

5.5.3 Tunneling

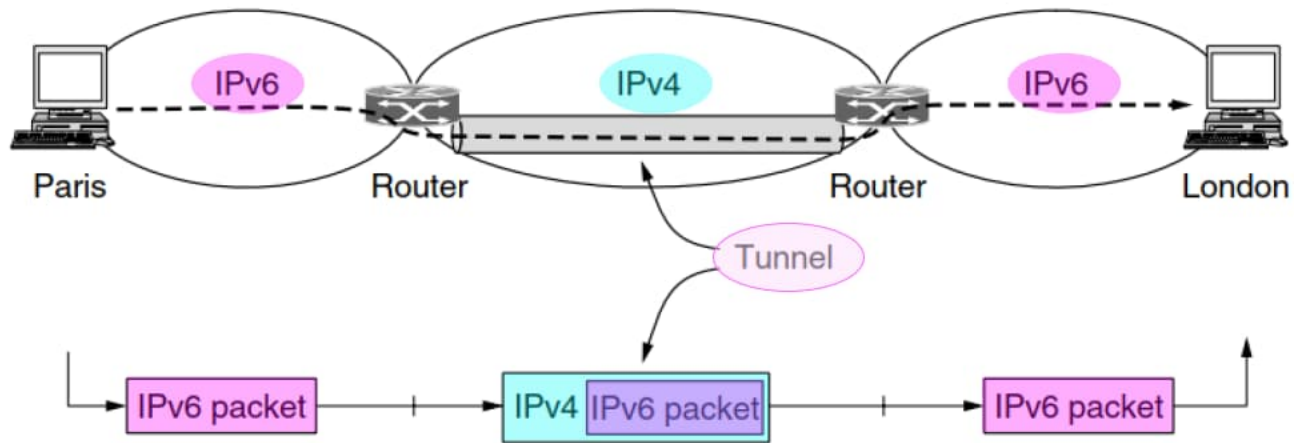


Figure 5-40. Tunneling a packet from Paris to London.

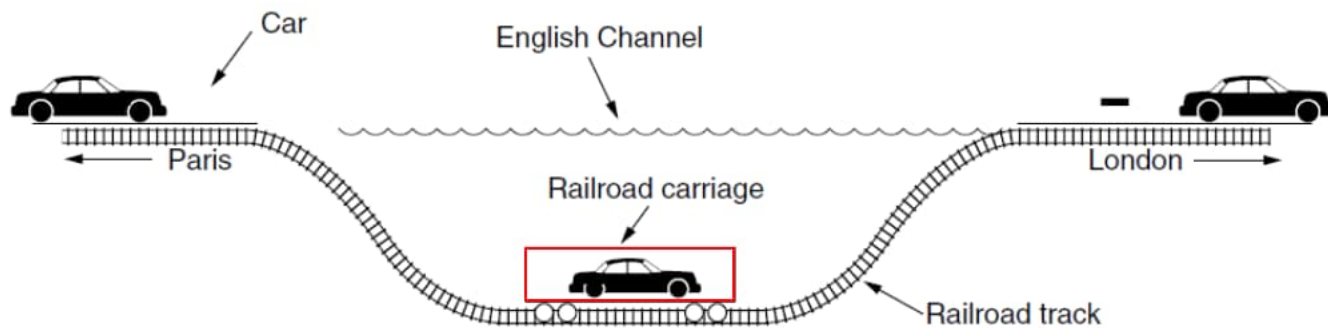


Figure 5-41. Tunneling a car from France to England.

5.5.4 Internetwork Routing

- Routing through an internet 互联网路由
 - ✦ poses the **same** basic problem as routing within a **single** network
 - ✦ but with some **added complications**.
 - networks may **internally** use **different routing algorithms**.
 - networks run by **different operators**.
- two-level routing algorithm 两级路由
 - ✦ **IGP: Interior Gateway Protocol** 内部网关路由协议
 - **RIP: Routing Information Protocol**
 - **OSPF: Open Shortest Path First**
 - ✦ **EGP: Exterior Gateway Protocol** 外部网关路由协议
 - **BGP: Border Gateway Protocol**
- **AS: Autonomous System** 自治系统

poses ['pəʊzɪz] v. 产生;引起; (问题)

5.5.5 Packet Fragmentation

➤ Why fragmentation

- ★ Each network or link imposes some **maximum size** on its packets. These limits have various causes, among them
 1. **Hardware** (e.g., the size of an Ethernet frame).
 2. **Operating system** (e.g., all buffers are 512 bytes).
 3. **Protocols** (e.g., the number of bits in the packet length field).
 4. Compliance with some (inter)national **standard**.
 5. Desire to **reduce** error-induced **retransmissions** to some level.
 6. Desire to **prevent** one packet from **occupying the channel** too long.

➤ MTU

- ★ **Maximum Transmission Unit** 最大传输单元
- ★ **path MTU** 路径最大传输单元

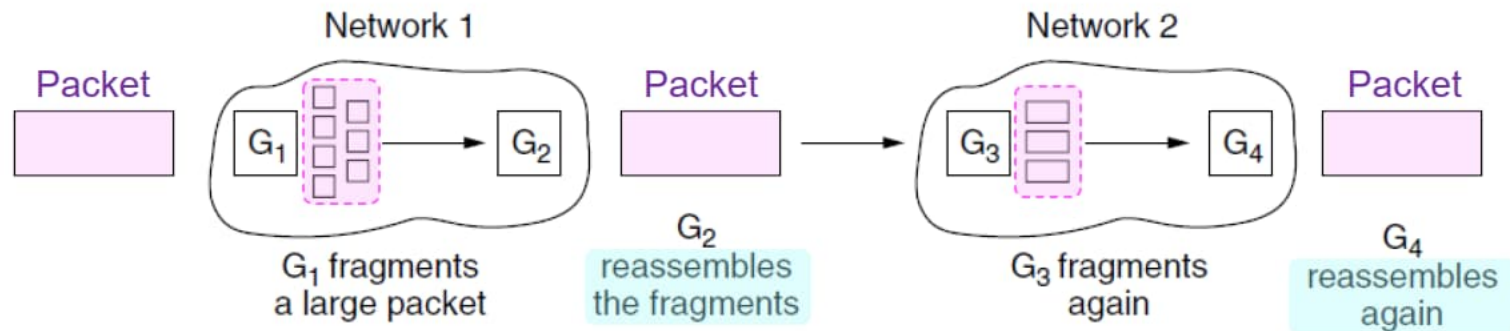
Fragmentation



- routers break up packets into fragments (路由器) 分片
- Two strategies
 - ✦ for recombining the fragments back into the original packet.
 - Transparent fragmentation 透明分片
 - Nontransparent fragmentation 非透明分片

Transparent fragmentation

- Each fragment is addressed to the **same exit router**
 - ✦ Subsequent networks are **not** even **aware** that fragmentation has occurred.
- problems
 - ✦ the **exit router** must **know** when it has received **all** the pieces
 - ✦ **all packets** must **exit via** the **same router**
 - ✦ the packet may **pass through a series of small** packet networks and need to be **repeatedly fragmented** and **reassembled**.



(a) Transparent fragmentation.

Nontransparent fragmentation

➤ Non-Transparent fragmentation

✦ reassembly is performed only at the destination host

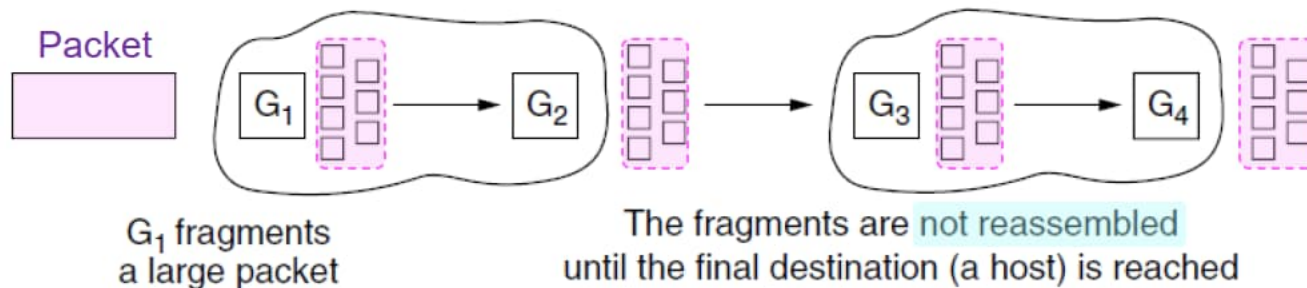
➤ advantage

✦ it requires routers to do less work

➤ problems

✦ The overhead can be higher, because fragment headers are now carried over some links

✦ a whole packet is lost if any of its fragments are lost



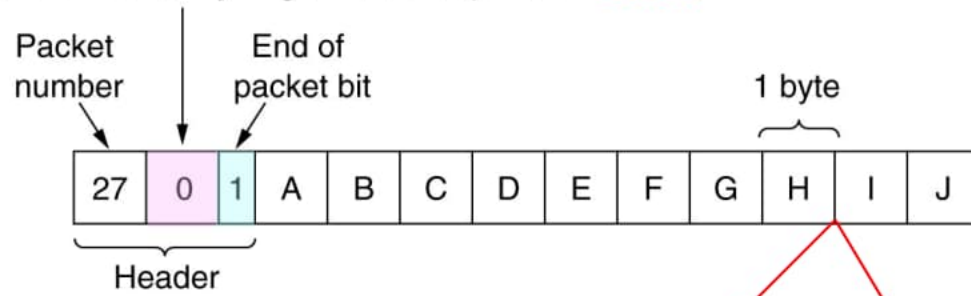
(b) Nontransparent fragmentation.

fragment offset

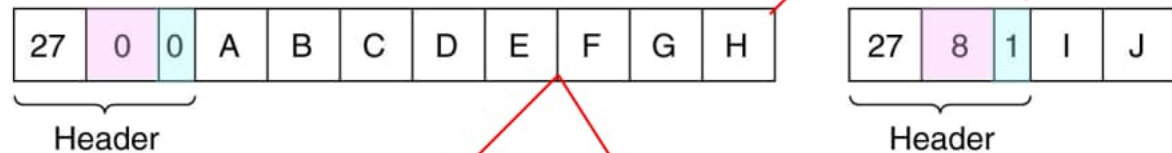


Number of the first elementary fragment in this packet **offset**

original



the 1st fragmentation



the 2nd times

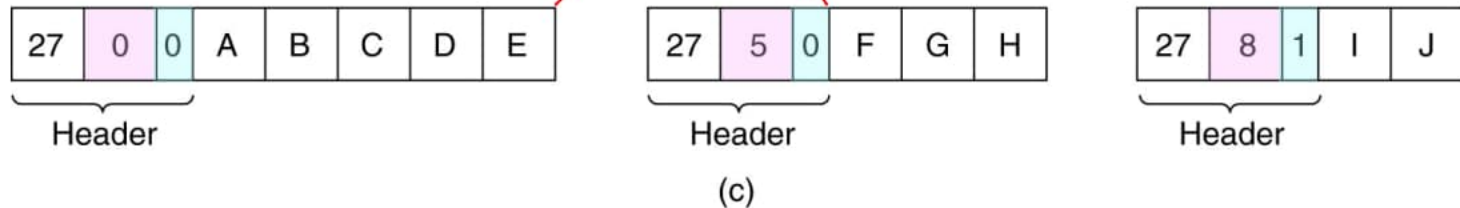


Figure 5-43. Fragmentation when the elementary data size is 1 byte.

(a) **Original** packet, containing 10 data bytes.

(b) Fragments after passing through a network with maximum packet size of 8 payload bytes plus header.

(c) Fragments after passing through a **size 5** gateway.

Path MTU discovery



路径MTU发现

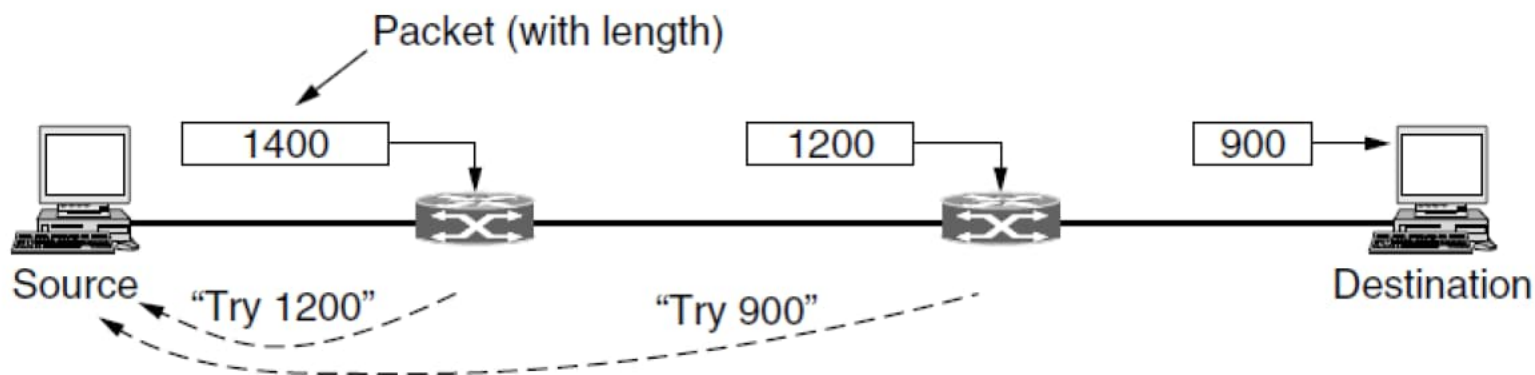


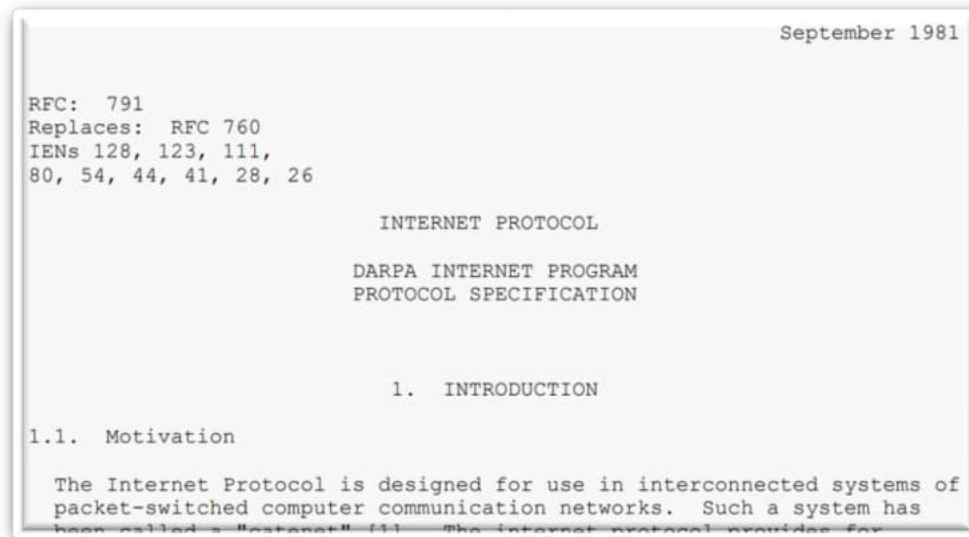
Figure 5-44. Path MTU discovery.



5.6 The Network Layer in the Internet



- The **network layer** in **the Internet** 因特网上的网络层
 - ✦ the **Internet** can be viewed as
 - a collection of **networks** or **ASes** (**Autonomous Systems**) that are interconnected.
 - ✦ The glue that **holds** the whole Internet together is
 - the network layer protocol, **IP** (**I**nternet **P**rotocol, **RFC791**).
 - ✦ <https://www.rfc-editor.org/rfc-index.html>



The Internet

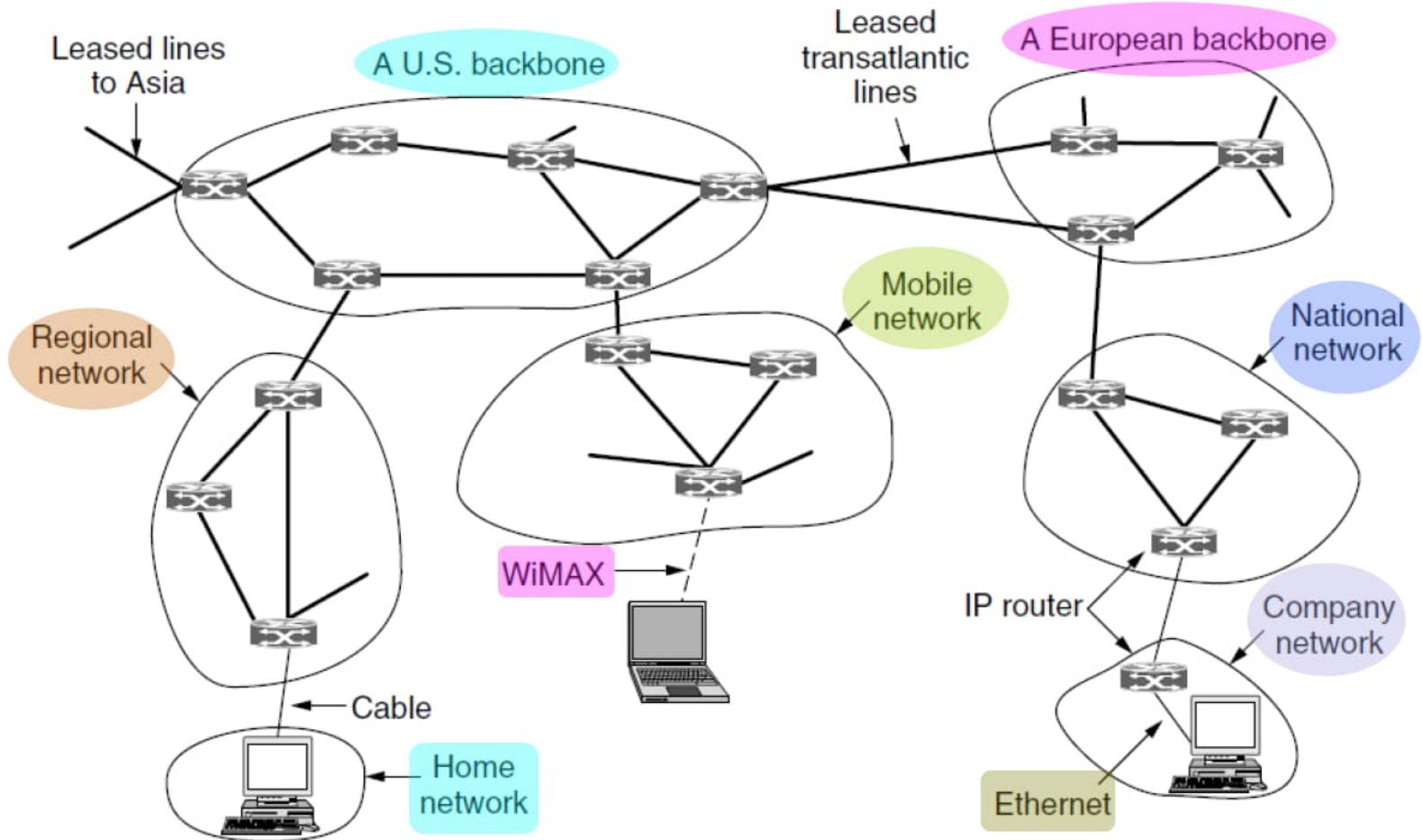


Figure 5-45. The Internet is an interconnected collection of many networks.

Design Principles for Internet



most
important



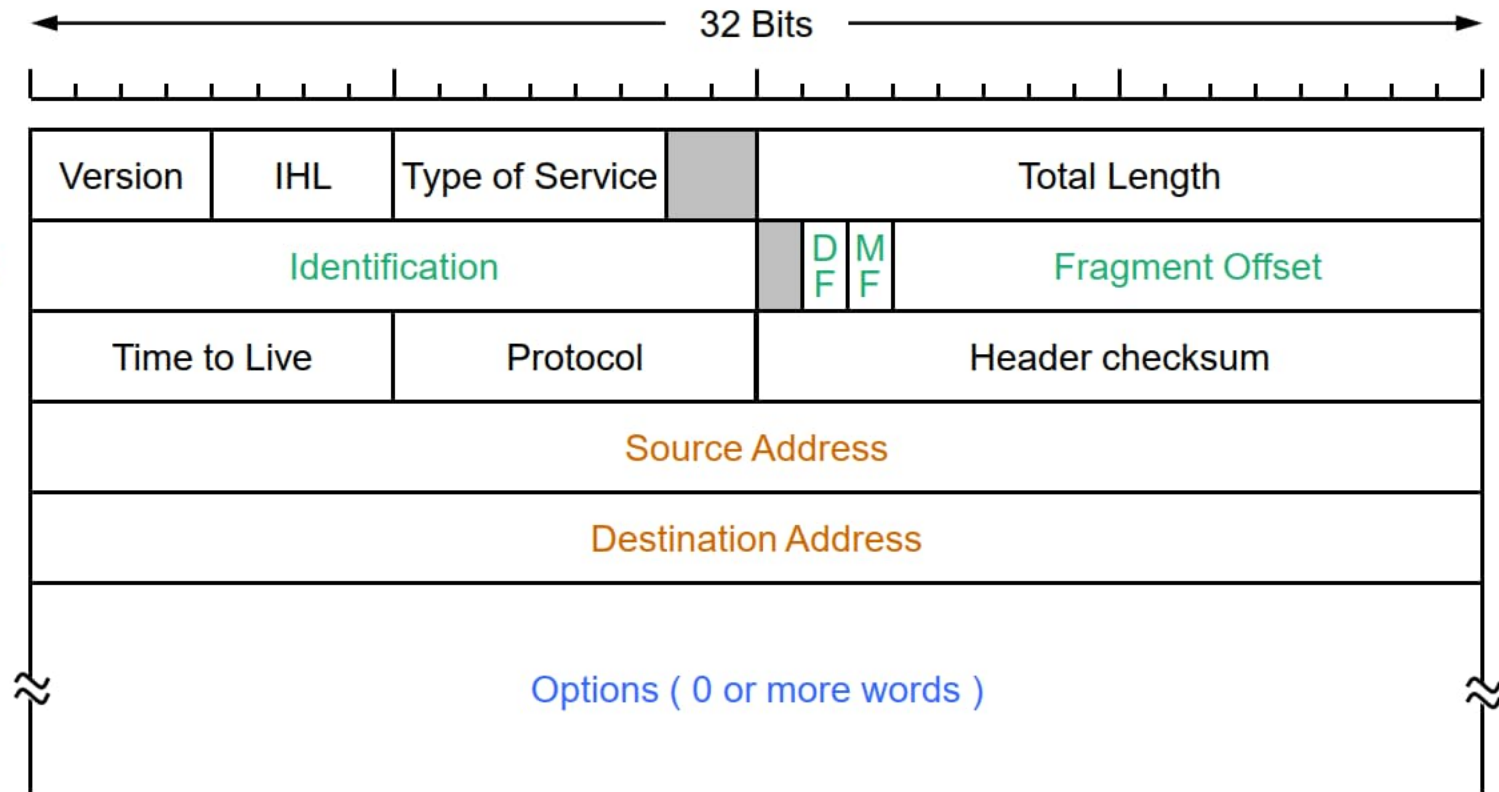
- Make sure it works.
- Keep it simple.
- Make clear choices.
- Exploit modularity.
- Expect heterogeneity.
- **Avoid** static options and parameters.
- Look for a good design; it need **not** be perfect.
- Be strict when sending and tolerant when receiving.
- Think about scalability.
- Consider performance and cost.

least
important

5.6.1 The IP Version 4 Protocol



分片控制



The IPv4 (Internet Protocol) **header**.

IP options



Option	Description
Security	Specifies how secret the datagram is
Strict source routing	Gives the complete path to be followed
Loose source routing	Gives a list of routers not to be missed
Record route	Makes each router append its IP address
Timestamp	Makes each router append its address and timestamp

Figure 5-47. Some of the IP options.

5.6.2 IP Addresses

➤ IP address

- ✦ can be used in the **Source** address and **Destination** address
- ✦ Each **32-bit** address is comprised of
 - a variable-length **network portion** in the top bits and 网络部分
 - a **host portion** in the bottom bits. 主机部分
- ✦ is written in **dotted decimal notation**, 点分十进制表示法
 - each of the 4 bytes is written in decimal, from 0 to 255.
 - For example, the 32-bit **hexadecimal** address **D22DD001**
 - is written as **210.45.208.1**

Network Prefix



➤ Network prefix 网络前缀

- ✦ a network corresponds to a **contiguous block** of IP address space. This block is called a **prefix**.
- ✦ notation: **ip_addr/L**
- ✦ for example: **210.45.208.1/22**

contiguous: 毗邻的, 相连的
continuous: 连续的, 持续的

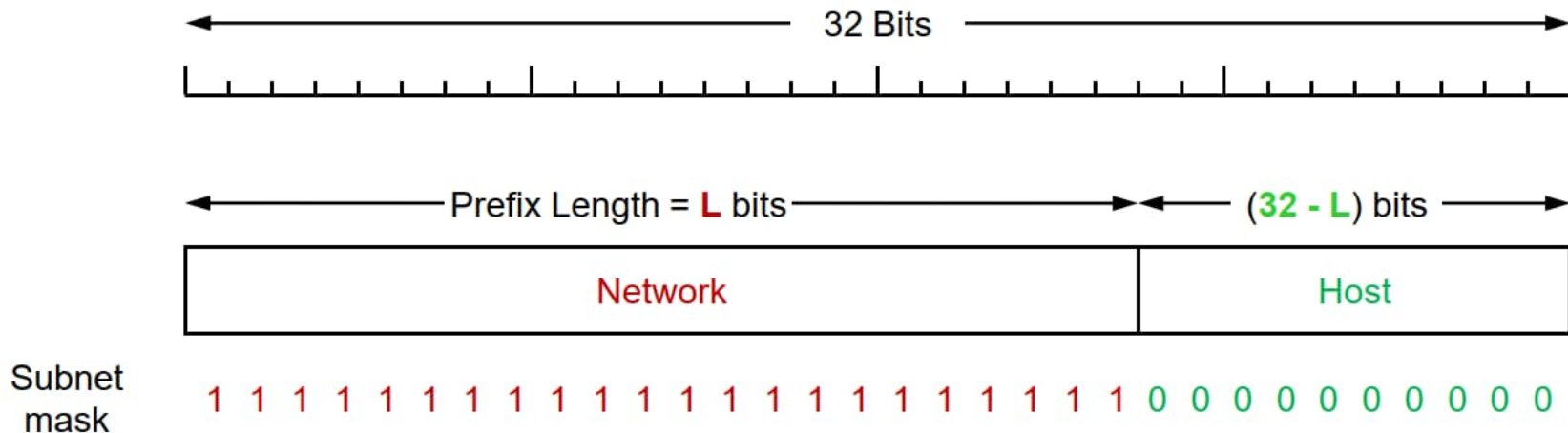


Figure 5-48. An IP prefix and a subnet mask.

Subnets



➤ Subnetting 子网

- ✦ the block of addresses is split into several parts
 - for internal use as multiple networks,
 - while still acting like a single network to the outside world.

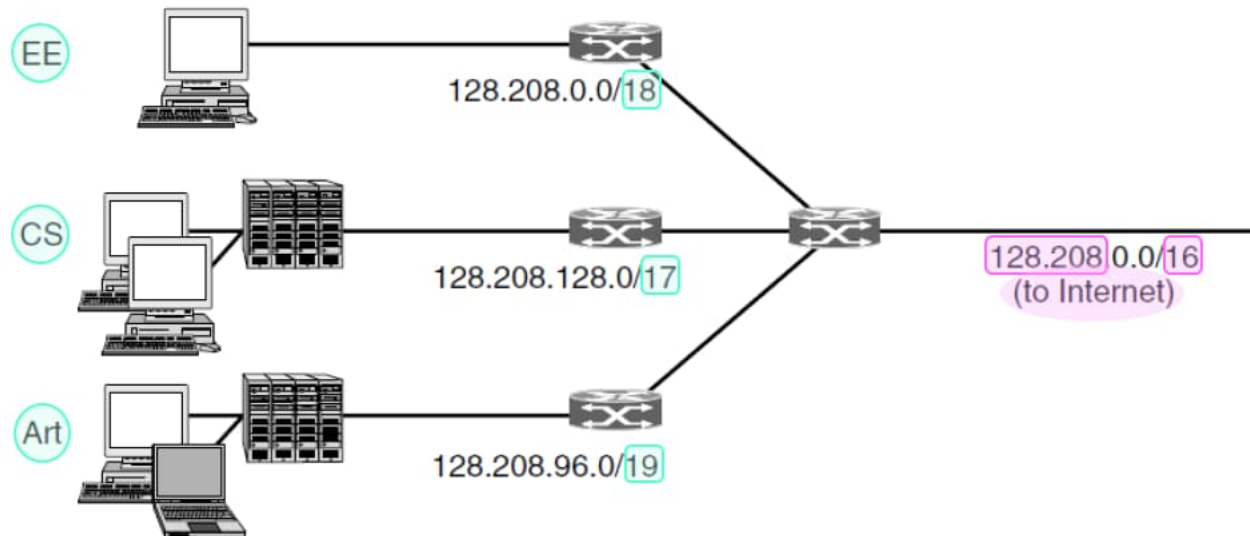
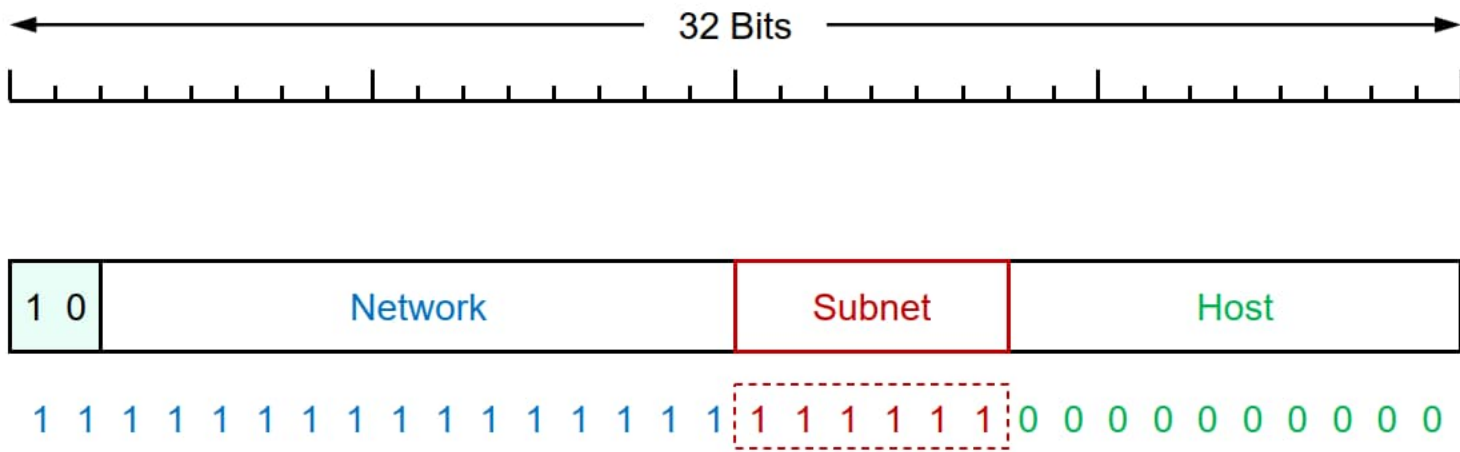


Figure 5-49. Splitting an IP prefix into separate networks with subnetting.

Subnet mask



子网掩码



A class B network subnetted into 64 subnets.

CIDR: Classless InterDomain Routing



- problem
 - ✦ routing table explosion
- route aggregation 路由聚合
 - ✦ combine multiple small prefixes into a single larger prefix
- CIDR - Classless InterDomain Routing 无类别域间寻经
 - ✦ RFC 4632

route: 路径, 路由
routing: 寻路, 寻径

Aggregation of IP prefixes

University	First address	Last address	How many	Prefix	
Cambridge	194.24.0.0	194.24.7.255	2048	194.24.0.0/21	0=0000 0000
Edinburgh	194.24.8.0	194.24.11.255	1024	194.24.8.0/22	8=0000 1000
(Available)	194.24.12.0	194.24.15.255	1024	194.24.12.0/22	12=0000 1100
Oxford	194.24.16.0	194.24.31.255	4096	194.24.16.0/20	16=0001 0000

16bits + 3bits = 19bits

Figure 5-50. A set of IP address assignments.

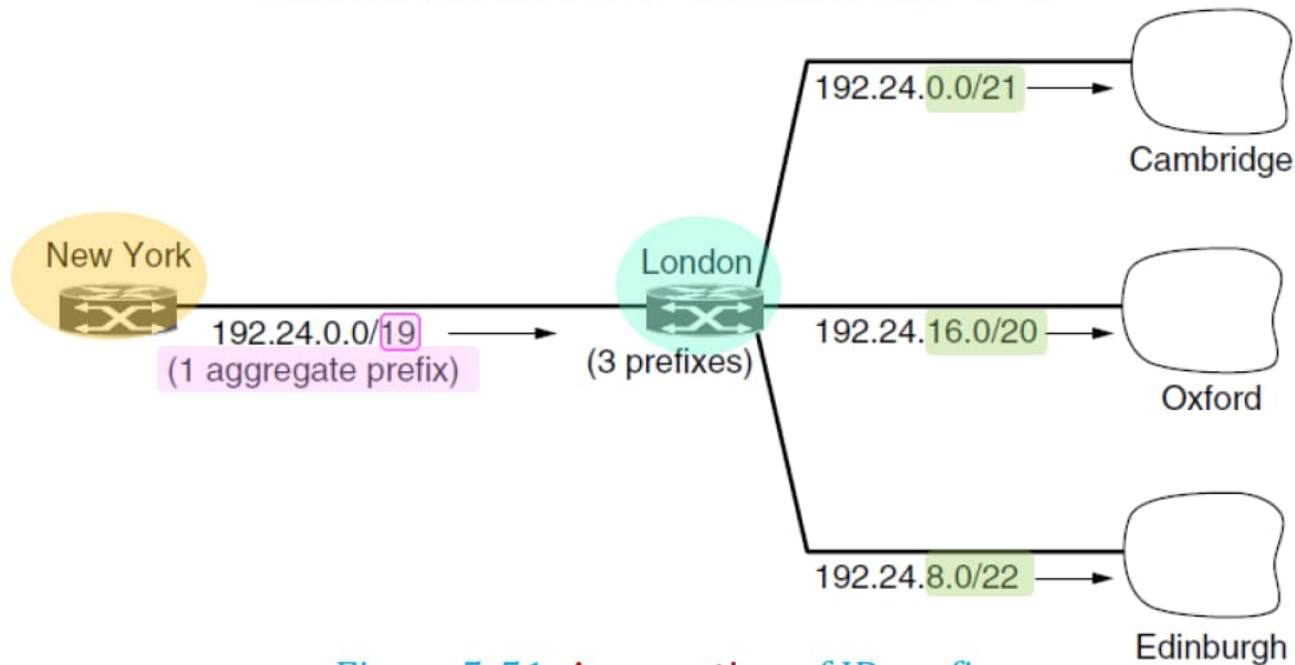


Figure 5-51. Aggregation of IP prefixes.

Longest matching prefix routing



最长匹配前缀路由

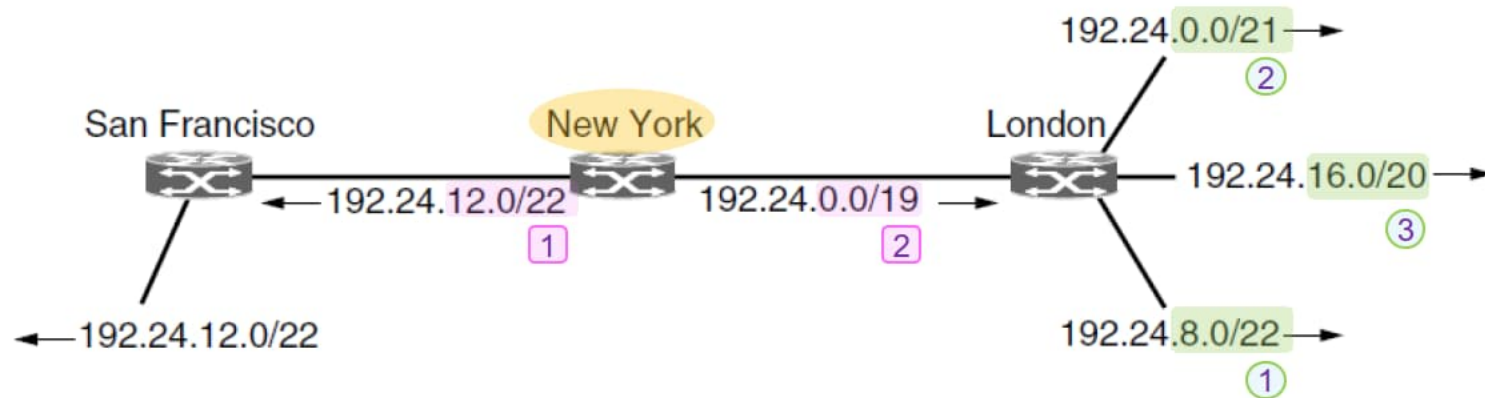


Figure 5-52. Longest matching prefix routing at the New York router.

Classful Addressing



全类别编址方案

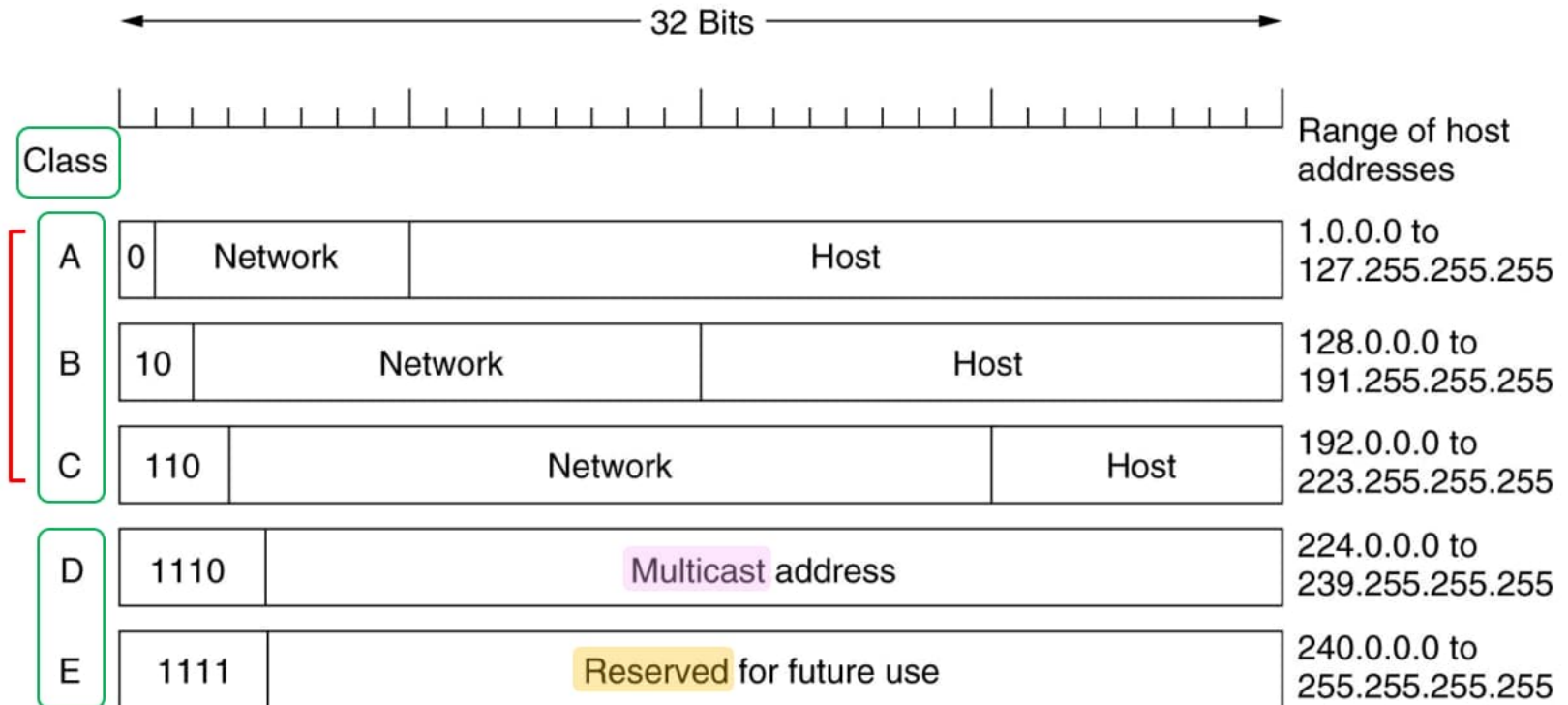


Figure 5-53. IP address formats.

address: 地址

addressing: 编址, 编址方案

Special IP addressing



0 0																																This host								
0 0								...								0 0								Host																A host on this network
1 1																																Broadcast on the local network								
Network								1 1 1 1								...								1 1 1 1								Broadcast on a distant network								
127								(Anything)																								Loopback								

Figure 5-54. Special IP addresses.

NAT



- NAT: Network Address Translation 网络地址翻译
- Intranet or Private Internet
- three reserved ranges, RFC1918
 - ★ 10.0.0.0 - 10.255.255.255, A single Class A network
 - ★ 172.16.0.0 - 172.31.255.255, 16 contiguous Class B networks
 - ★ 192.168.0.0 - 192.168.255.255, 256 contiguous Class C networks

NAT box

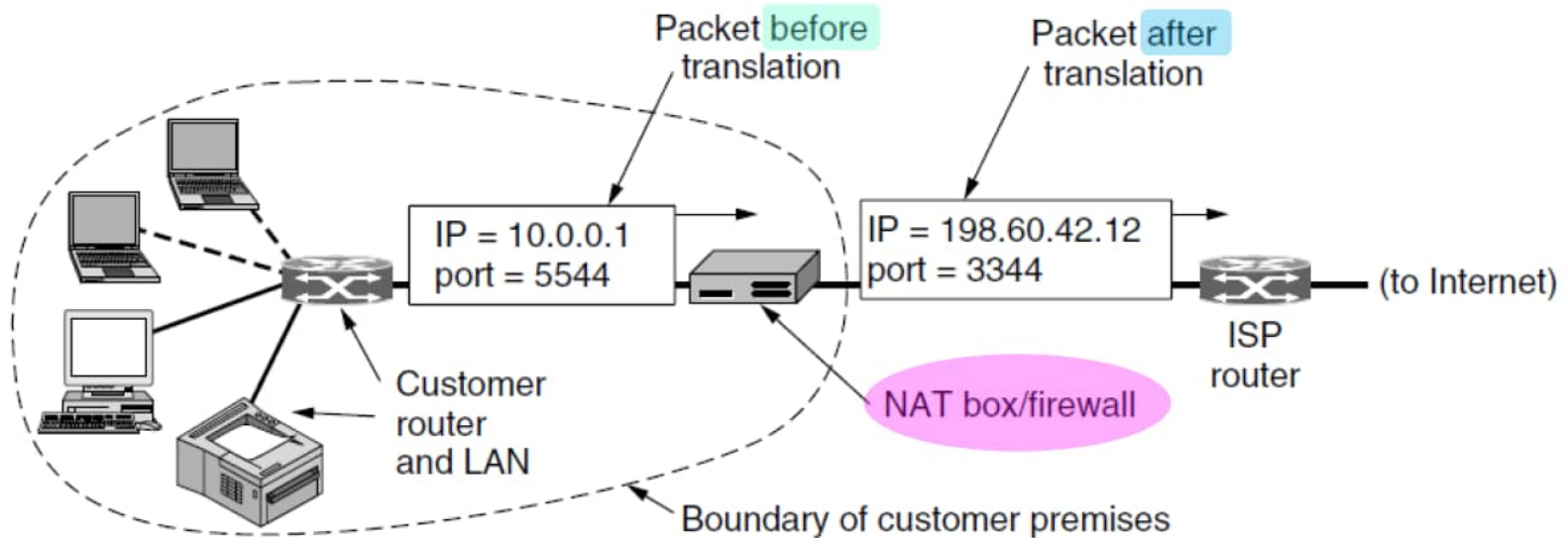


Figure 5-55. Placement and operation of a NAT box.

5.6.3 IP Version 6



➤ Protocol Background

- ✦ IP has been **patched** (subnets, supernets) but there is still the fundamental **32 bits** address limitation
- ✦ IETF started effort to specify new version of IP in **1991**
 - New version would require **change of header**
 - Include **all modifications** in **one new protocol**
 - Solicitation of suggestions from **community**
 - Result was IP**ng** which became **IPv6**
 - First version completed in **1994**
- ✦ Same **architectural** principles as v4 – only bigger

ng: next generation

➤ RFC 8200, July 2017

- ✦ Internet Protocol, Version 6 (**IPv6**) Specification

major goals of IPv6



1. Support **billions of hosts**, even with inefficient address allocation.
2. **Reduce** the size of the **routing tables**.
3. **Simplify** the **protocol**, to allow routers to process packets **faster**.
4. Provide **better security** (authentication and privacy).
5. Pay more attention to **the type of service**, particularly for **real-time data**.
6. Aid **multicasting** by allowing **scopes** to be specified.
7. Make it possible for a host to **roam** without changing its address.
8. Allow the protocol to **evolve** in the future.
9. Permit the old and new protocols to **coexist** for years.

What Ever Happened to IPv5?

0	IP	March 1977 version	(deprecated)
1	IP	January 1978 version	(deprecated)
2	IP	February 1978 version A	(deprecated)
3	IP	February 1978 version B	(deprecated)
4	IPv4	September 1981 version	(current widespread)
5	ST	Stream Transport	(not a new IP, little use)
6	IPv6	December 1998 version	(formerly SIP, SIPP)
7	CATNIP	IPng evaluation	(formerly TP/IX; deprecated)
8	Pip	IPng evaluation	(deprecated)
9	TUBA	IPng evaluation	(deprecated)
10-15		(unassigned)	

SIPP (Simple Internet Protocol Plus)

IPv6 planned support list



- 128-bit address space
 - ✦ This is what it's all about ...
- Real-time / QoS services
- Security and authentication
- Autoconfiguration
 - ✦ Hosts autoconfig with IP address and domain name
 - ✦ Idea is to try to make systems more plug-n-play
- Enhanced routing functionality eg. Mobile hosts
- Multicast
- Protocol extensions
- Smooth transition path from IPv4

Why was 128 bits chosen as the IPv6 address size?



- Proposals for **fixed**-length, **64-bit** addresses
 - ✦ Accommodates 10^{12} sites, 10^{15} nodes, at **.0001** allocation efficiency
 - ✦ **Minimizes** growth of per-packet header **overhead**
 - ✦ **Efficient** for software processing on current CPU hardware
- Proposals for **variable**-length, up to **160 bits**
 - ✦ Compatible with deployed OSI NSAP addressing plans
 - ✦ Accommodates auto-configuration using IEEE 802 addresses
 - ✦ Sufficient structure for projected number of service providers
- **Settled** on **fixed**-length, **128-bit** addresses
 - ✦ (340,282,366,920,938,463,463,374,607,431,768,211,456 **in all**!)

IPv6 Technology Scope



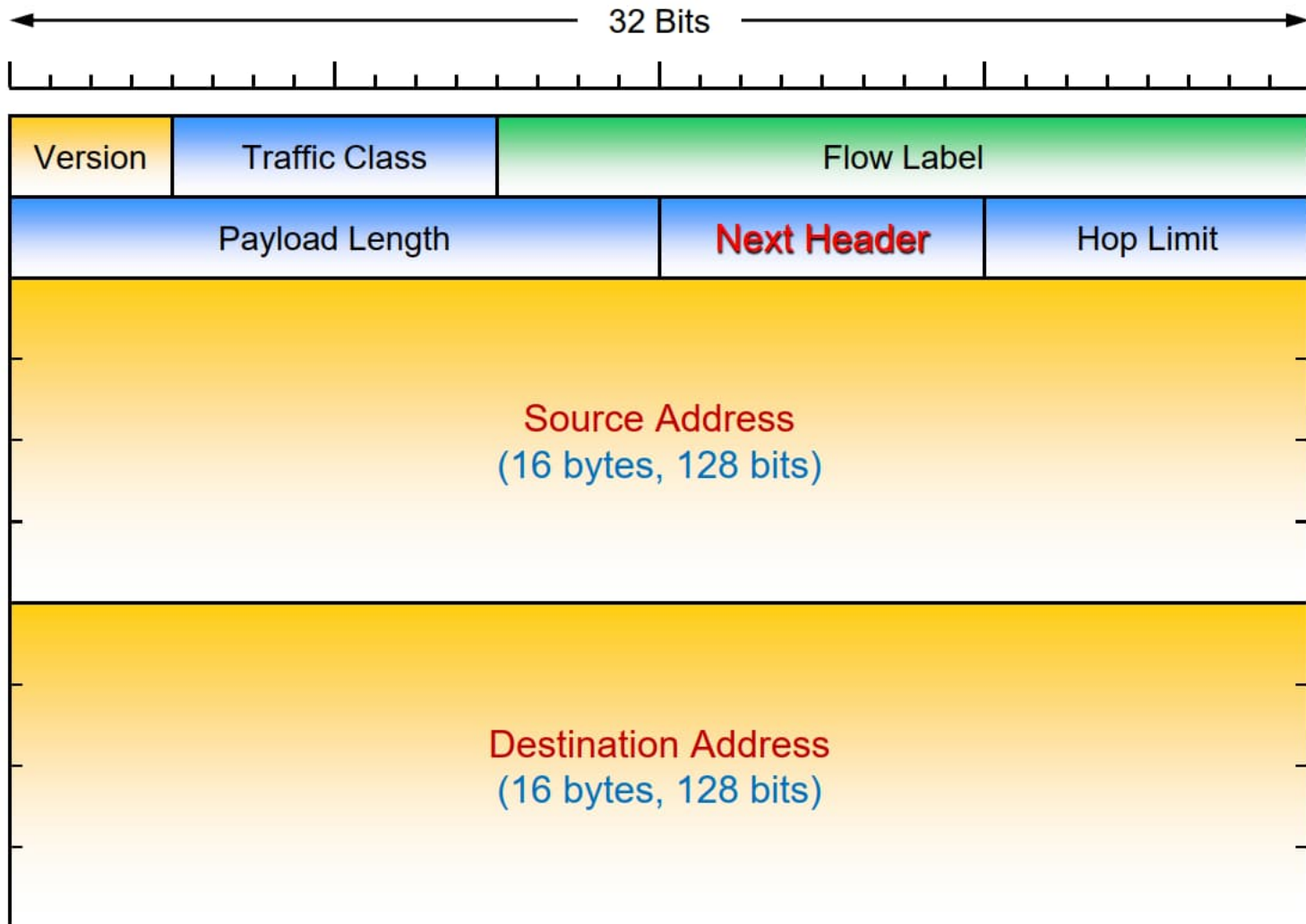
IP Service	IPv4 Solution	IPv6 Solution
Addressing Range	32-bit, Network Address Translation	128-bit, Multiple Scopes
Autoconfiguration	DHCP	Serverless, Reconfiguration, DHCP
Security	IPSec	IPSec Mandated, works End-to-End
Mobility	Mobile IP	Mobile IP with Direct Routing
Quality-of-Service	Differentiated Service, Integrated Service	Differentiated Service, Integrated Service
IP Multicast	IGMP/PIM/Multicast BGP	MLD/PIM/Multicast BGP, Scope Identifier

Main IPv6 Benefits

- Expanded **addressing** capabilities
- Structured hierarchy to manage **routing table** growth
- **Serverless** autoconfiguration and reconfiguration
- **Streamlined header** format and **flow** identification
- Improved support for options / extensions
- **Security** - Built-in, strong IP-layer encryption and authentication
- **Mobility** - More efficient and robust mechanisms
- **Quality of Service** (**flow label**)
- Privacy Extensions for **Stateless** Address Autoconfiguration
- Source address selection

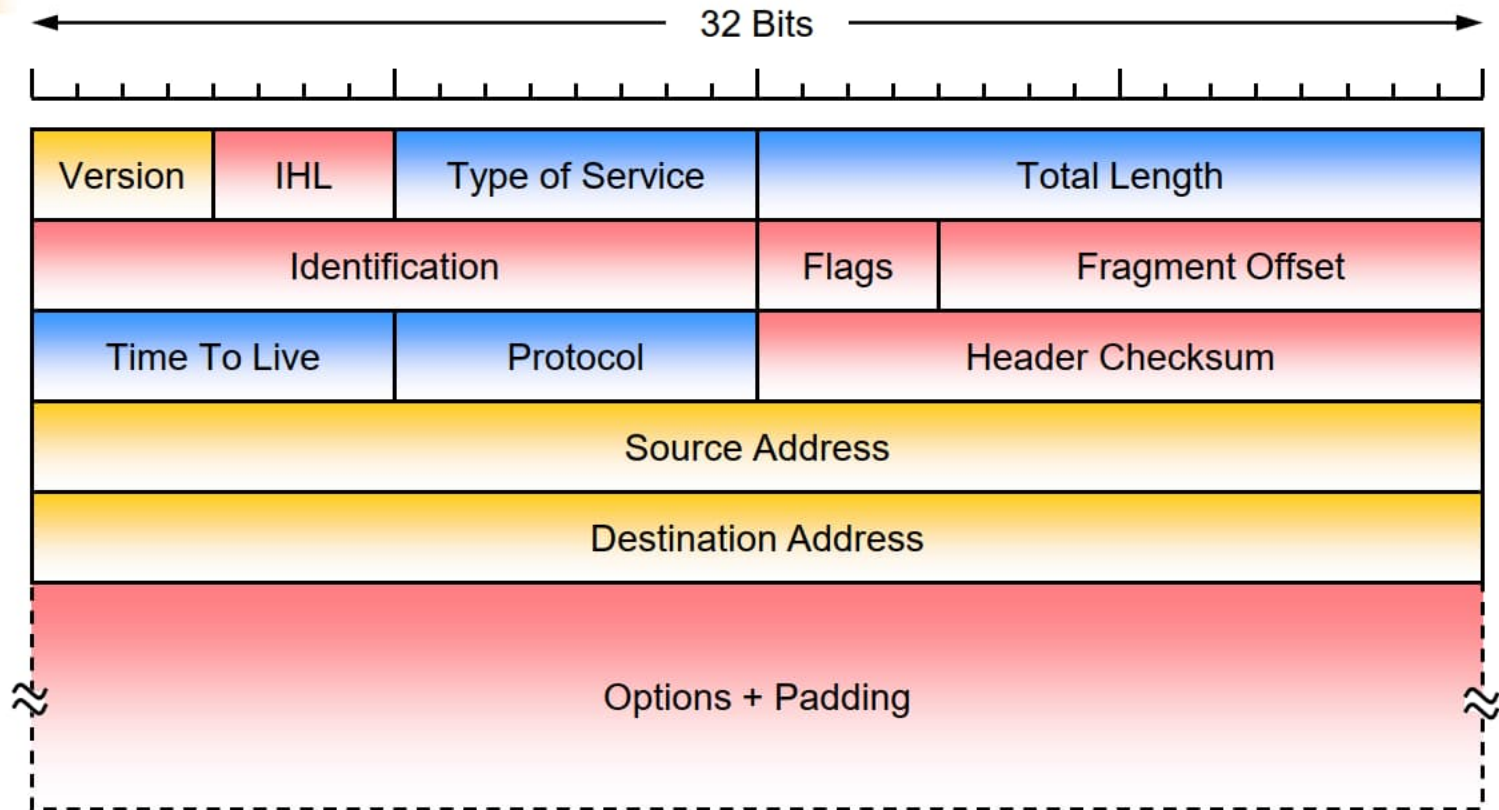
streamline *vt.* 使...更高效

The Main IPv6 Header



IPv6 Base Header

IPv4 Header



- Legend**
-  - field's name kept from IPv4 to IPv6
 -  - fields not kept in IPv6
 -  - Name & position changed in IPv6
 -  - New field in IPv6

Summary of Header Changes between IPv4 & IPv6



➤ Streamlined

- ✦ Fragmentation fields moved out of base header
- ✦ IP options moved out of base header
- ✦ Header Checksum eliminated
- ✦ Header Length field eliminated
- ✦ Length field excludes IPv6 header
- ✦ Alignment changed from 32 to 64 bits

➤ Revised

- ✦ Time to Live → Hop Limit
- ✦ Protocol → Next Header
- ✦ Precedence & TOS → Traffic Class
- ✦ Addresses increased 32 bits → 128 bits

➤ Extended

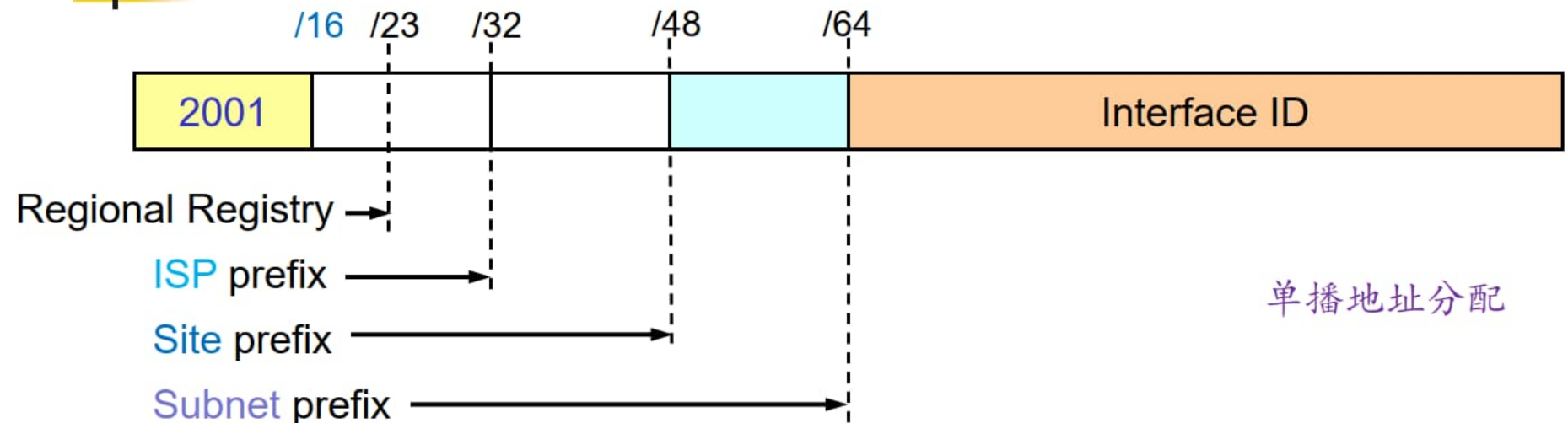
- ✦ Flow Label field added

Text Representation of Addresses



- RFC 4291, Feb. 2006
 - ✦ IP Version 6 Addressing Architecture
 - ✦ Updated by RFC5952, RFC6052, RFC7136, RFC7346, RFC7371, RFC8064
- eight groups of four hexadecimal digits with colons between the groups
 - ✦ "Preferred" form: 1080:0:FF:0:8:800:200C:417A
 - 首选
 - ✦ Compressed form: FF01:0:0:0:0:0:0:43
 - 缩写
 - becomes FF01::43
 - ✦ IPv4-mapped: 0:0:0:0:0:FFFF:10.1.68.3
 - 映射IPv4
 - or ::FFFF:10.1.68.3
 - ✦ IPv4-compatible: 0:0:0:0:0:0:10.1.68.3
 - 兼容IPv4
 - or ::10.1.68.3

Address Allocation



- The allocation process was recently updated by the registries:
 - ✦ IANA allocates from 2001::/16 to regional registries
 - ✦ Each regional registry allocation is a ::/23
 - ✦ ISP allocations from the regional registry is a ::/36 (immediate allocation) or ::/32 (initial allocation) or shorter with justification
 - ✦ Policy expectation that an ISP allocates a ::/48 prefix to each customer

IANA: The Internet Assigned Numbers Authority
因特网号码分配机构

Extension Headers



Extension header	Description
Hop-by-hop options 逐跳选项头标	Miscellaneous information for routers
Routing 信源路由头标	Loose list of routers to visit
Fragmentation 分片头标	Management of datagram fragments
Authentication 认证头标	Verification of the sender's identity
Encapsulating security payload 封装化安全净荷头标	Information about the encrypted contents
Destination options 信宿选项头标	Additional information for the destination

Figure 5-57. IPv6 extension headers.

Jumbo payload header



特大净荷头标

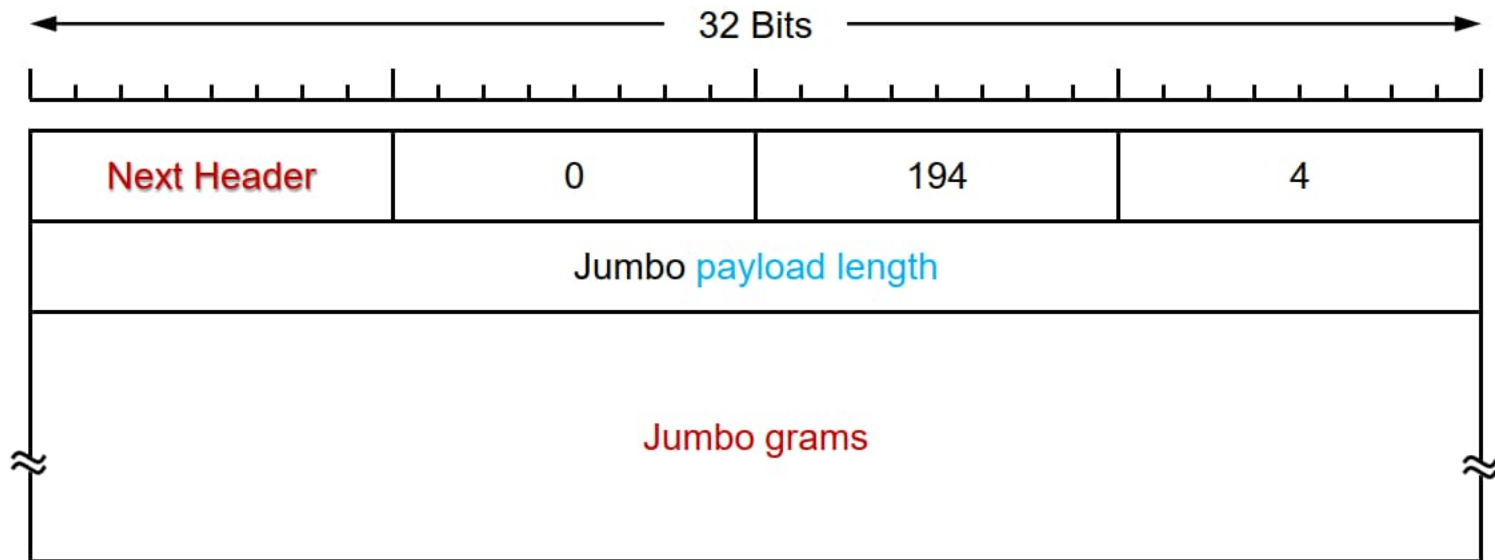


Figure 5-58. The hop-by-hop extension header for large datagrams (jumbograms).

- jumbograms 特大净荷
 - ✦ datagrams exceeding 64KB

Routing header



信源路由头标

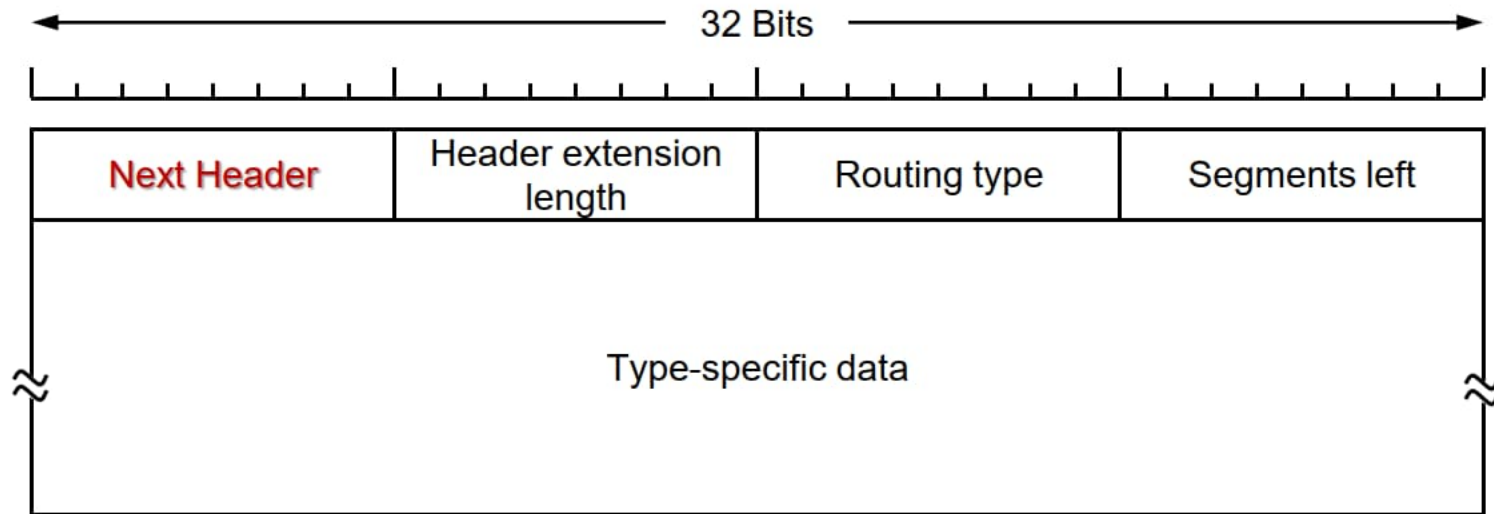
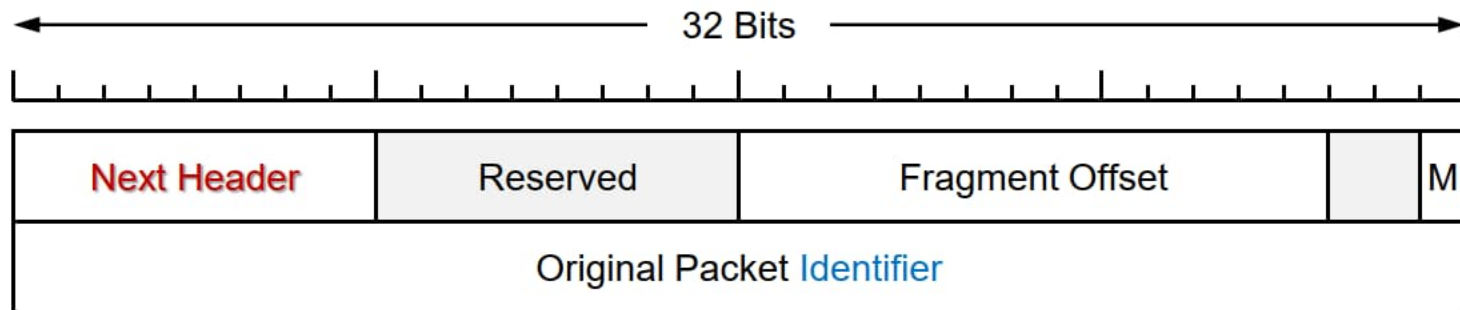


Figure 5-59. The extension header for routing.

Fragment Header



分片头标



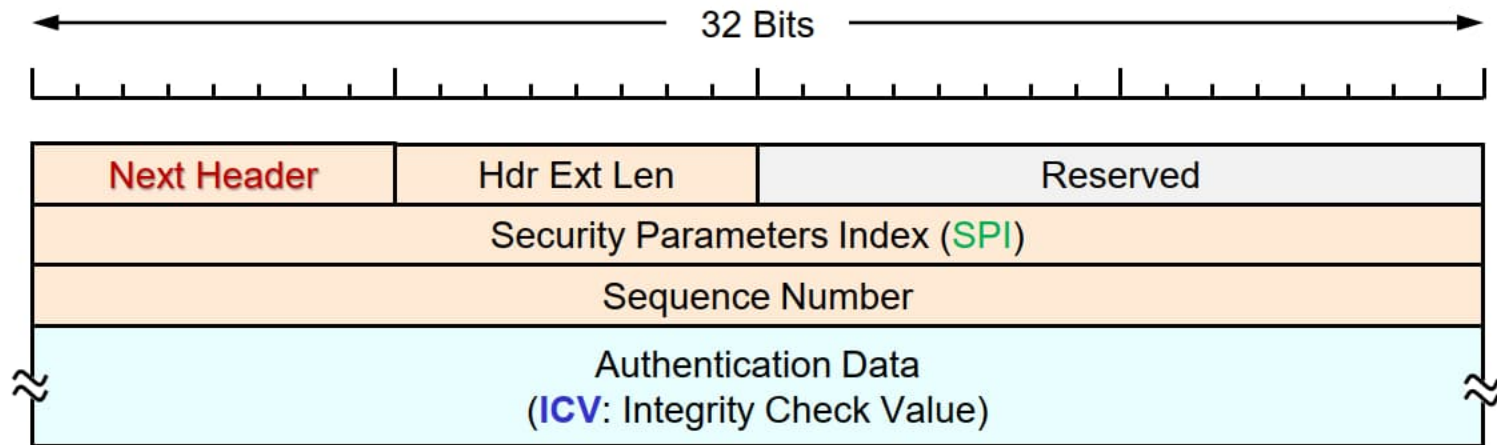
- IPv6 fragmentation & reassembly is an **end-to-end** function;
routers do not fragment packets
 - ✦ if packet is too big they send ICMP "packet too big"
- though **discouraged**, can use IPv6 Fragment header to support **upper layers that do not (yet) do path MTU discovery**

Authentication Header



RFC 4302

认证头标



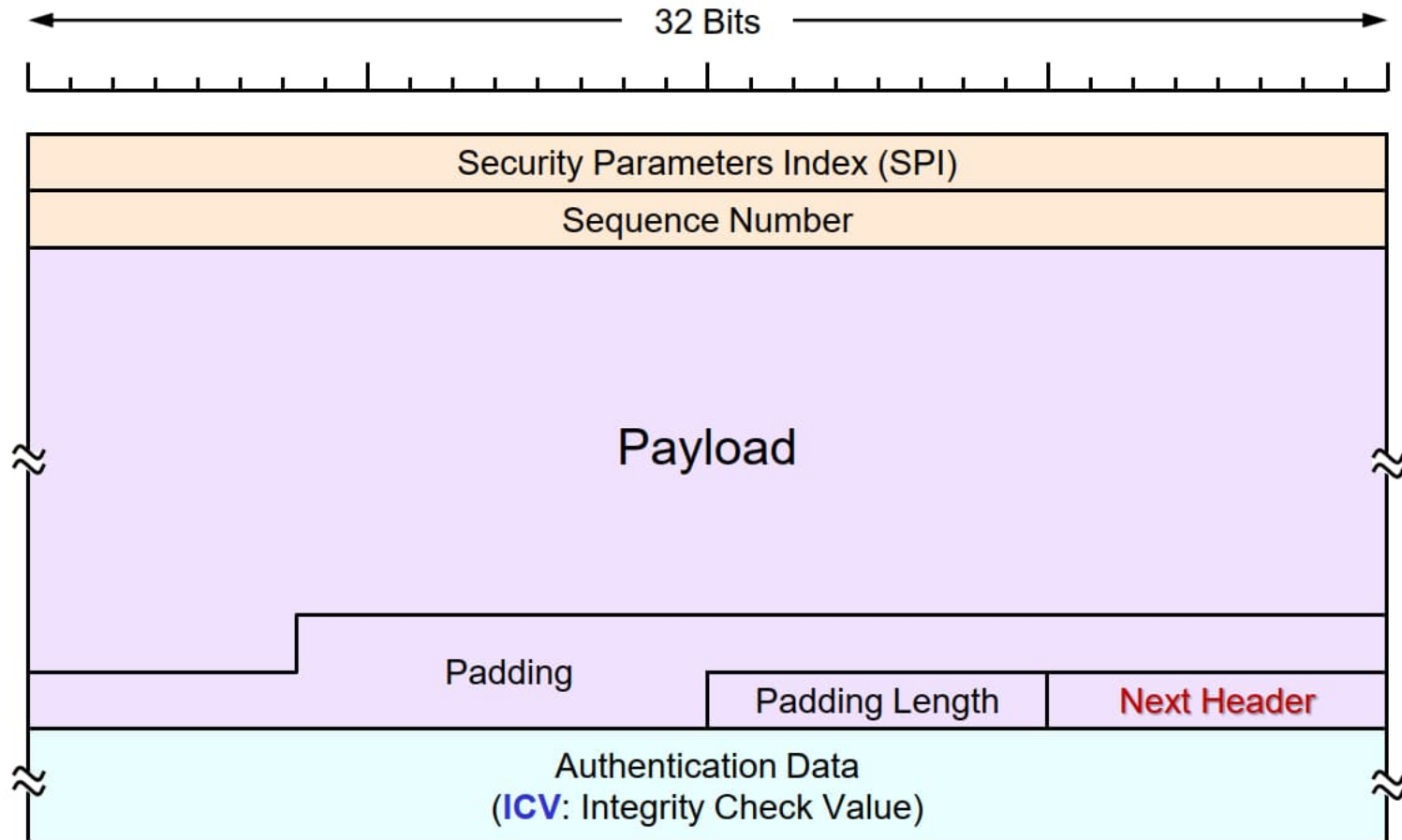
- Destination Address + SPI identifies SA (Security Association) state (key, lifetime, algorithm, etc.)
- Provides authentication and data integrity for all fields of IPv6 packet that do not change en-route
- Default algorithm is Keyed MD5

Encapsulating Security Payload (ESP)



RFC 4303

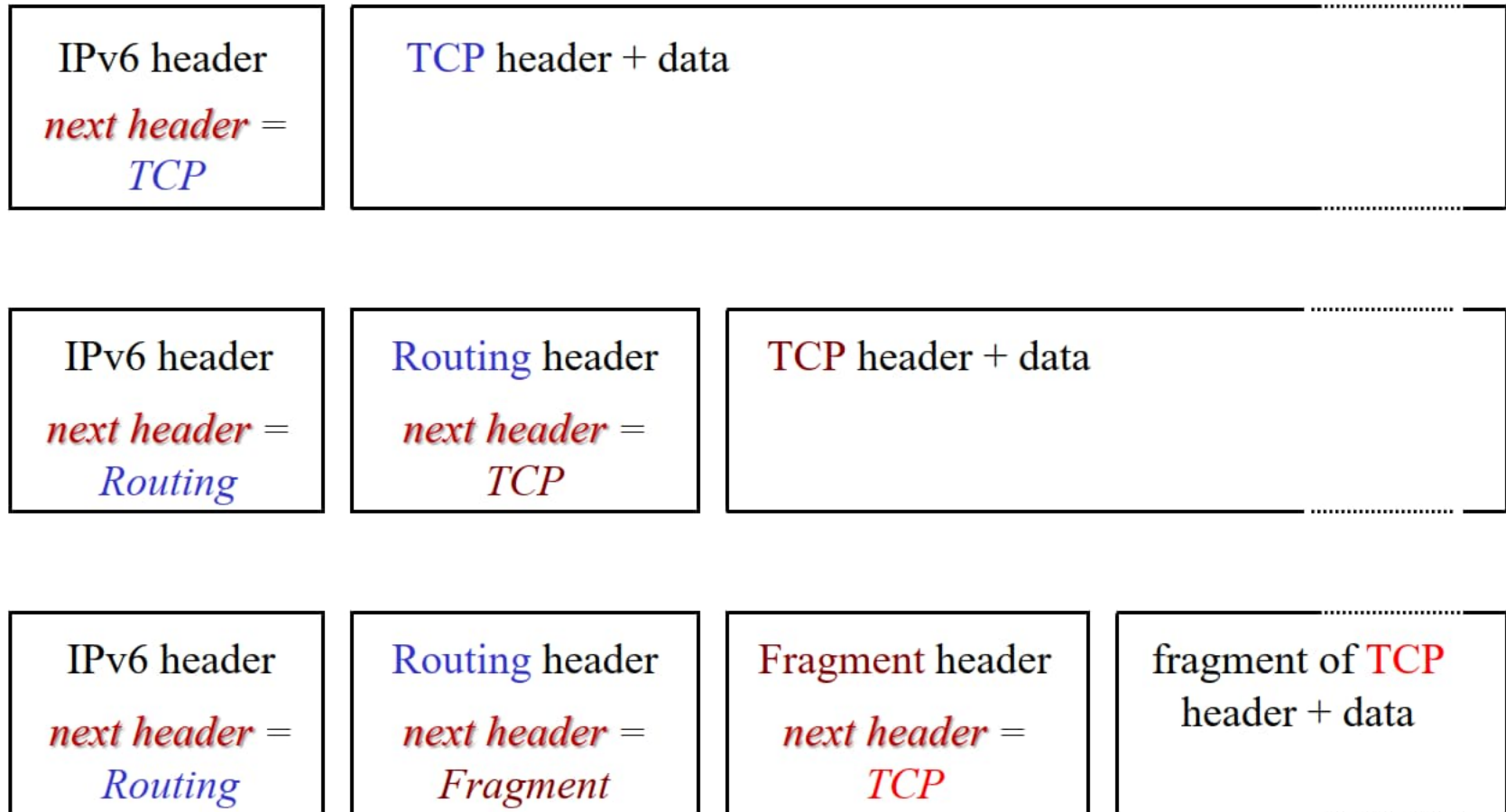
封装化安全净荷



Chains of Extension Headers



链式头标扩展



Controversies



- address length
 - ✦ 16-byte fixed-length addresses
- the length of the Hop limit field
 - ✦ limiting the maximum number of hops to 255 (8-bit field)
- maximum packet size
 - ✦ normal packets are limited to 64 KB, but the hop-by-hop extension header can be used to permit jumbograms.
- removing the IPv4 checksum
 - ✦ IPv6 does not have a checksum.
- Mobile hosts
- security

5.6.4 Internet Control Protocols



➤ ICMP: Internet Control Message Protocol

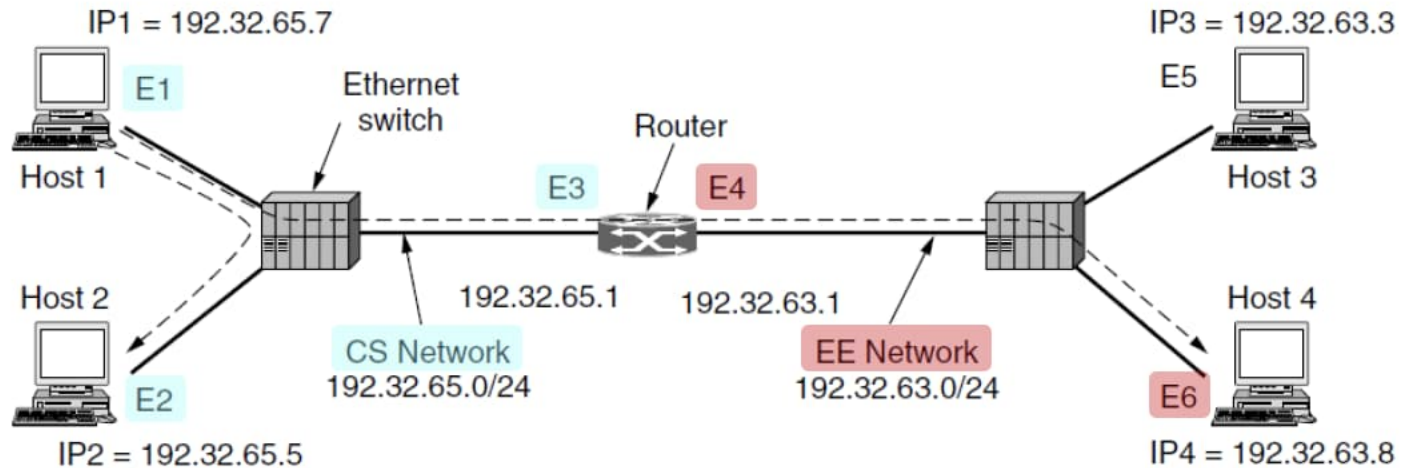
✦ 网际(互联网)控制报文协议

Message type	Description
Destination unreachable	Packet could not be delivered
Time exceeded	Time to live field hit 0
Parameter problem	Invalid header field
Source quench	Choke packet
Redirect	Teach a router about geography
Echo and echo reply	Check if a machine is alive
Timestamp request / reply	Same as Echo, but with timestamp
Router advertisement / solicitation	Find a nearby router

ARP



➤ ARP: Address Resolution Protocol 地址解析协议



Frame	Source IP	Source Eth.	Destination IP	Destination Eth.
Host 1 to 2, on CS net	IP1	E1	IP2	E2
Host 1 to 4, on CS net	IP1	E1	IP4	E3
Host 1 to 4, on EE net	IP1	E4	IP4	E6

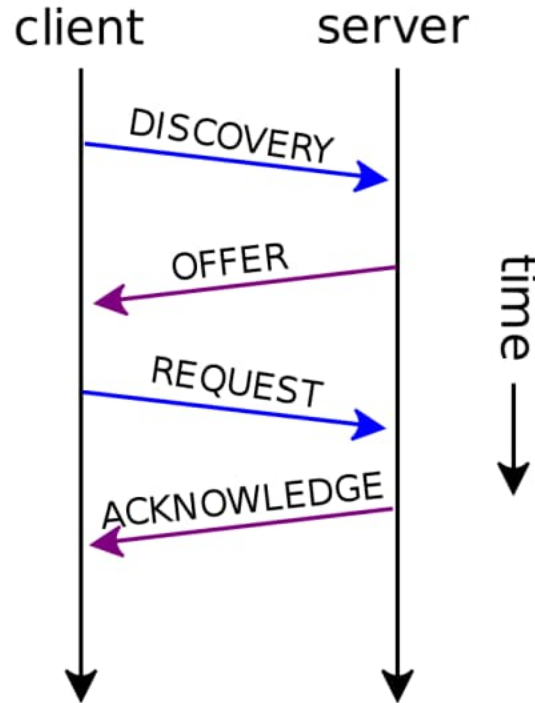
Figure 5-61. Two switched Ethernet LANs joined by a router.

DHCP



➤ DHCP: Dynamic Host Configuration Protocol

✦ 动态主机配置协议



5.6.5 Label Switching and MPLS

- LS: **Label Switching** 标记交换
- ✦ MPLS: **Multi-Protocol Label Switching** 多协议标记交换

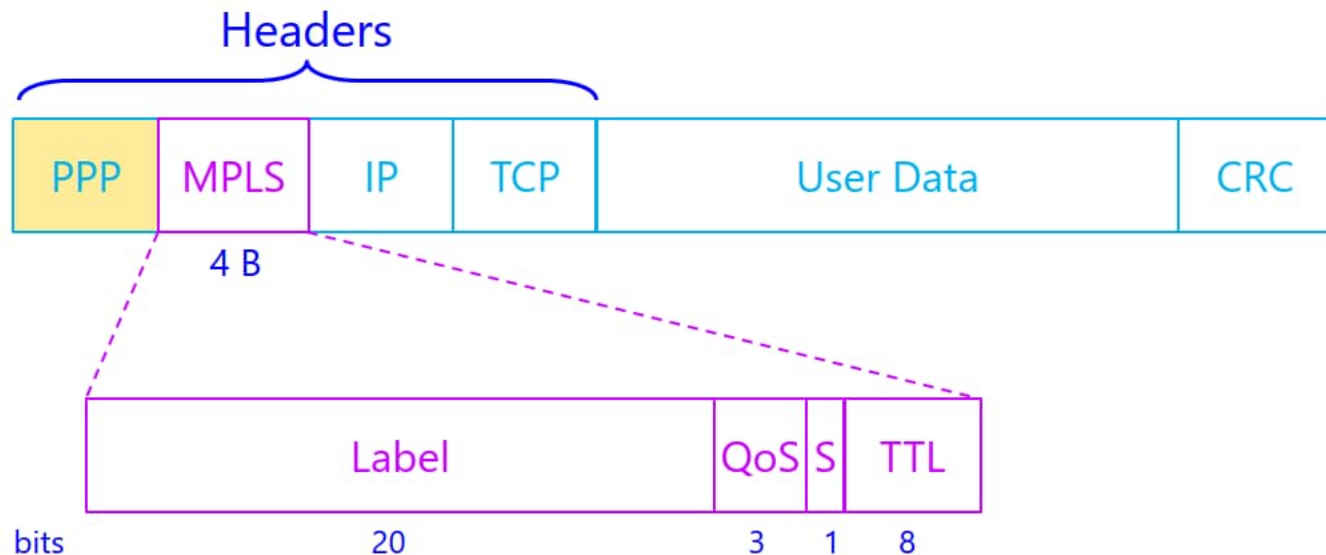


Figure 5-62. Transmitting a TCP segment using IP, MPLS, and PPP.

S bit: stack of labels

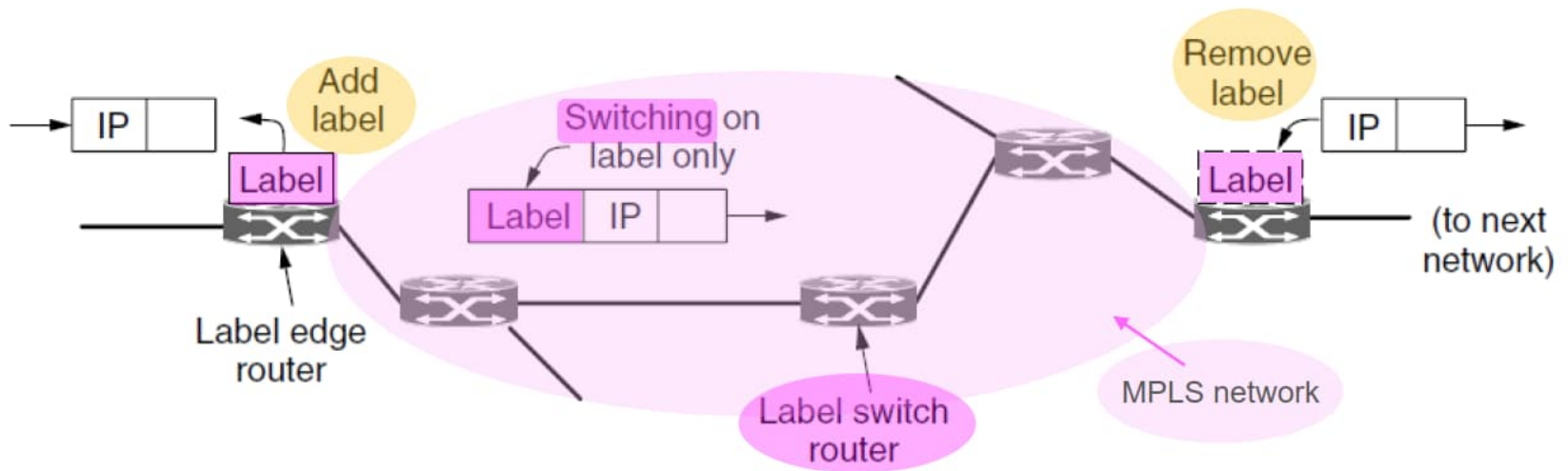


Figure 5-63. Forwarding an IP packet through an MPLS network.

5.6.6 OSPF—An Interior Gateway Routing Protocol

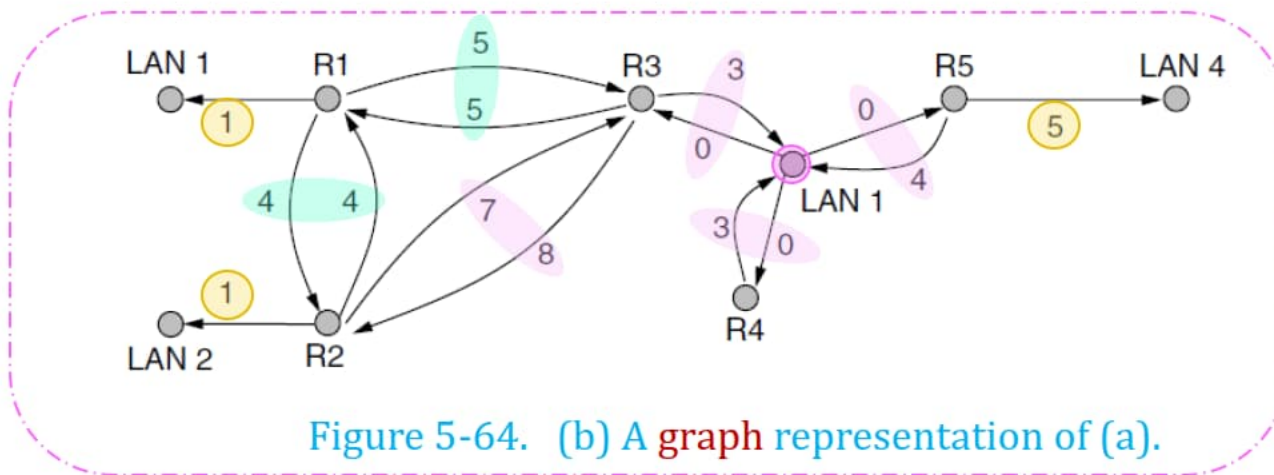
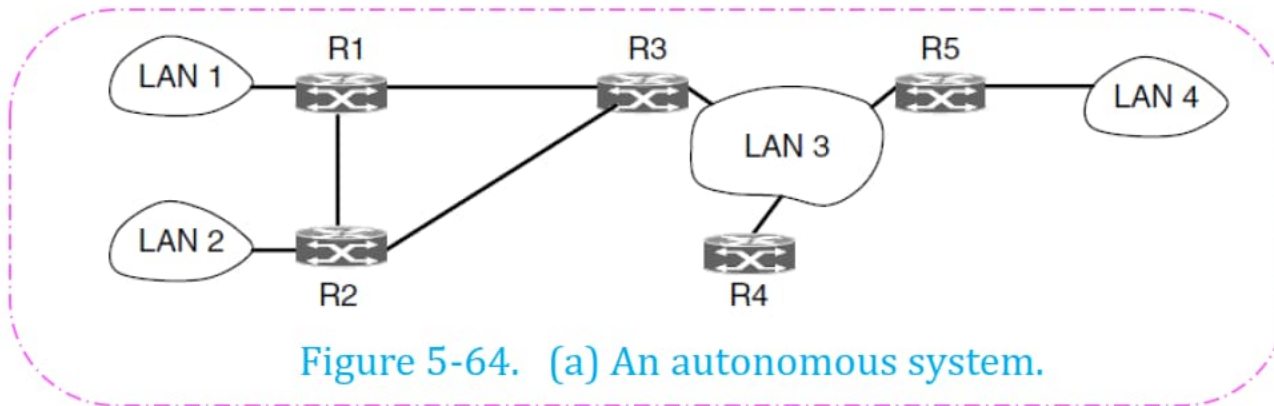


- **AS: Autonomous System** 自治系统
 - ✦ independent networks
 - ✦ operated by different organizations
 - usually a company, university, or ISP
 - ✦ use its own algorithm for internal routing, or intradomain routing
- **IGP: Interior Gateway Protocol** 内部网关协议
 - ✦ An intradomain routing protocol
 - for example: RIP, OSPF
- **EGP: Exterior Gateway Protocol** 外部网关协议
 - ✦ An interdomain routing protocol
 - for example: BGP

- **OSPF: Open Shortest Path First** 开放最短路径优先
 - ✦ the algorithm had to be published in the **open** literature;
 - ✦ the new protocol had to **support** a **variety** of **distance metrics**,
 - including **physical distance**, **delay**, and so on;
 - ✦ it had to be a **dynamic** algorithm;
 - ✦ it had to **support** routing based on **type of service**
 - ✦ OSPF had to do **load balancing**, splitting the load over multiple lines.
 - ✦ **support** for **hierarchical** systems
 - ✦ some modicum of **security**
 - to prevent fun-loving students from spoofing routers by sending them false routing information.
 - ✦ dealing with **routers** that were connected to the Internet **via a tunnel**

graph representation

- ✦ different weights in different direction
- ✦ ECMP: Equal Cost MultiPath 等代价-多路径(并发转发)



Four classes of routers

border router: 边界路由器
boundary router: 边缘路由器

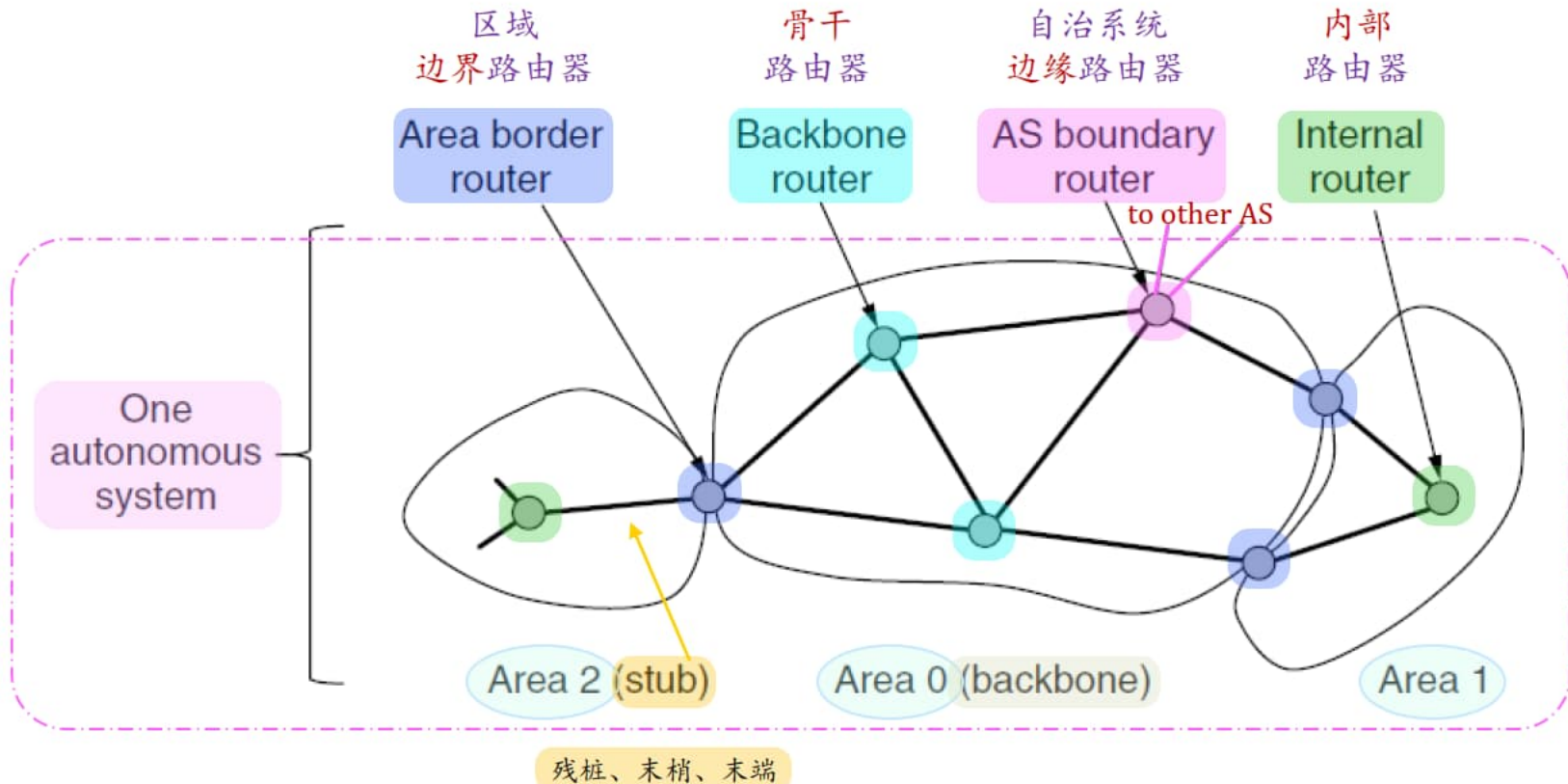


Figure 5-65. The relation between ASes, backbones, and areas in OSPF.

stub: 存根, 插桩

Five types of OSPF messages



Message type	Description
Hello	Used to discover who the neighbors are
Link state update	Provides the sender's costs to its neighbors
Link state ack	Acknowledges link state update
Database description	Announces which updates the sender has
Link state request	Requests information from the partner

Figure 5-66. The five types of OSPF messages.

5.6.7 BGP—The Exterior Gateway Routing Protocol

- **BGP: Border Gateway Protocol** 边界网关协议
 - ✦ used between **ASes**
 - ✦ **EGP: The Exterior Gateway Routing Protocol** 外部网关路由协议
 - **path vector protocol** 路径矢量协议

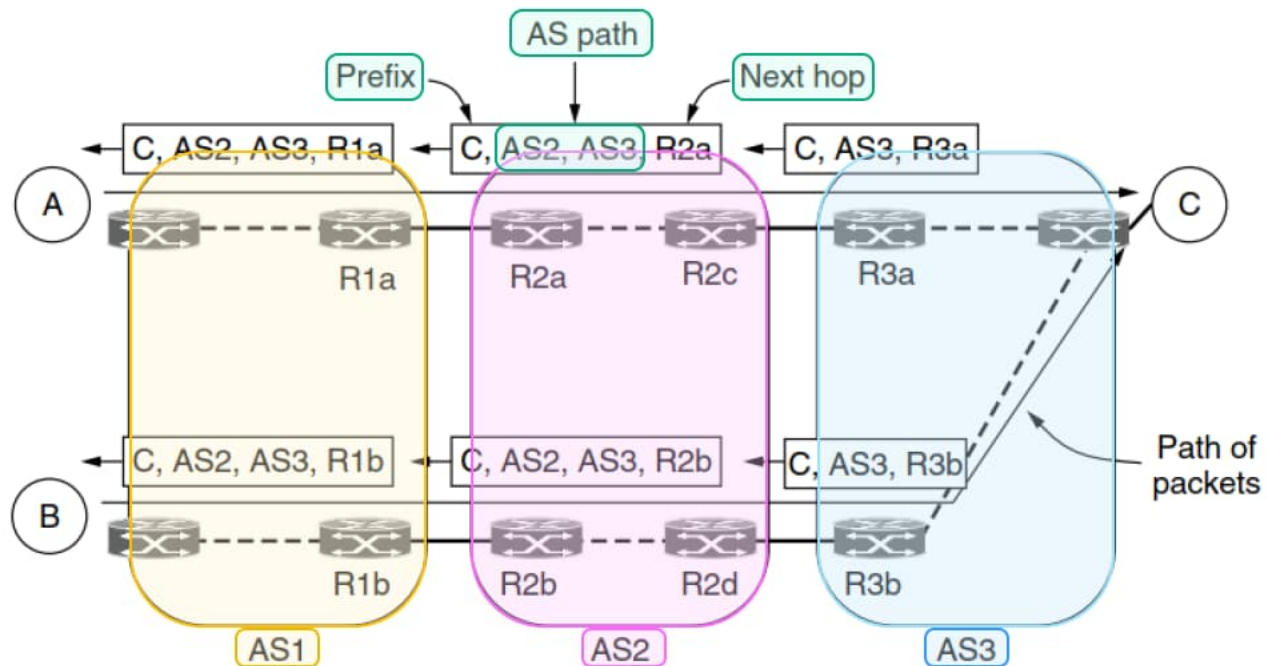


Figure 5-68. Propagation of BGP route advertisements.

Summary



- The **network** layer **provides services** to the **transport** layer.
 - ✦ **Connectionless**, **datagram**, a routing decision is made on every packet.
 - ✦ **Connection-oriented**, **virtual-circuit**, a routing is made when the virtual circuit is set up.
- Routing algorithm
 - ✦ **Flooding**, **distance (path) vector**, **link state**
 - ✦ **hierarchy** routing in large networks
- Congestion Control
- Quality of Service
 - ✦ throughput, delay, jitter, loss
- Protocols used in **the Internet**
 - ✦ IP (IPv6), ICMP, ARP, DHCP, MPLS
 - ✦ OSPF, BGP

Chapter 7 Application	
Chapter 6 Transport	
Chapter 5 Network	
Chapter 3 LLC	Data Link
Chapter 4 MAC	
Chapter 2 Physical	

